

Joint 1D inversion of TEM and MT data and 3D inversion of MT data in the Hengill area, SW Iceland

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ABSTRACT

An extensive study of the resistivity structure of the Hengill area in SW Iceland was carried out by the combined use of TEM and MT soundings. Joint inversion of the collected data can correct for static shifts in the MT data, which can be severe due to large near-surface resistivity contrasts. Joint 1D inversion of 148 TEM/MT sounding pairs and a 3D inversion of a 60 sounding subset of the MT data were performed. The 3D inversion was based on full MT impedance tensors previously corrected for static shift. Both inversion approaches gave qualitatively similar results, and revealed a shallow resistivity layer reflecting conductive alteration minerals at temperatures of 100–240 °C. They also delineated a deep conductor at 3–10 km depth. The reason for this deep-seated high conductivity is not fully understood. The distribution of the deep conductors correlates with a positive residual Bouguer gravity anomaly, and with transform tectonics inferred from seismicity. One model of the Hengill that is consistent with the well temperature data and the deep conductor that does not attenuate S-waves, is a group of hot, solidified, but still ductile magmatic intrusions that are closely associated with the heat source for the geothermal system.

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1. Introduction

The greater Hengill geothermal region in SW Iceland (Fig. 1) is one of the largest high-temperature geothermal areas in the country. Large-scale utilization of geothermal resources started in 1990, when Reykjavík Energy (OR) commissioned a 100 MWt hot water plant at Nesjavellir, north of Mount Hengill, for space heating of the city of Reykjavík and surrounding communities. Since then, hot water production has been increased to 290 MWt.

Electricity generation at Nesjavellir started in 1998 when a 60 MWe power plant came online. Production was expanded in 2001 (30 MWe) and in 2005 (30 MWe) to the present (end of 2009) capacity of 120 MWe. In 2006, the first phase of the Hellisheiði power plant, built and operated by OR south of Mount Hengill, was commissioned with an installed capacity of 90 MWe. The Hellisheiði plant was expanded in 2007 (33 MWe) and 2008 (90 MWe) and is now generating 213 MWe. In other words, the total installed power production capacity in the Hengill geothermal area is 333 MWe.

Geothermal exploration in the Hengill region had been intermittently going on from the 1940s until the 1980s when it was intensified in preparation for the Nesjavellir plant. A brief overview of the exploration history is given in Section 3.

The multinational research initiative “Integrated Geophysical Exploration Technologies for deep fractured geothermal systems” (I-GET) was funded by the European Union in 2006. Its aim was to improve and further develop geophysical methods to explore for deep geothermal resources. The project consortium was led by the GeoForschungsZentrum, Potsdam in Germany and included Freie Universität Berlin and Geothermie Neubrandenburg GmbH in Germany; Enel Green Power, Instituto di Geoscienze e Georisorse and the University of Pisa in Italy; Scientific and Technical Centre (BRGM) in France, GEOWAT AG in Switzerland; the Centre for Renewable Energy in Greece; the Polish Academy of Sciences; and the Iceland GeoSurvey (ÍSOR). Four geothermal test sites in different geological settings were investigated as part of the project; i.e. Gross Schönebeck (Germany) and Skierniewice (Poland) in sedimentary rocks, Travale (Italy) in metamorphic rocks and Hengill (Iceland) in volcanic rocks.

ÍSOR led the Hengill effort and focused on the advanced use of electromagnetic (EM) methods for geothermal exploration. This group also did a magnetotelluric (MT) survey to supplement pre-existing EM data. In addition, BRGM carried out a broad band passive seismic survey (Jousset et al., 2010). This paper presents the results of the work carried out by ÍSOR in the Hengill area under the I-GET project.

2. Geological and tectonic setting

The Hengill volcanic complex is located in the southern end of the western volcanic zone (WVZ) of Iceland (Fig. 1). It is at the

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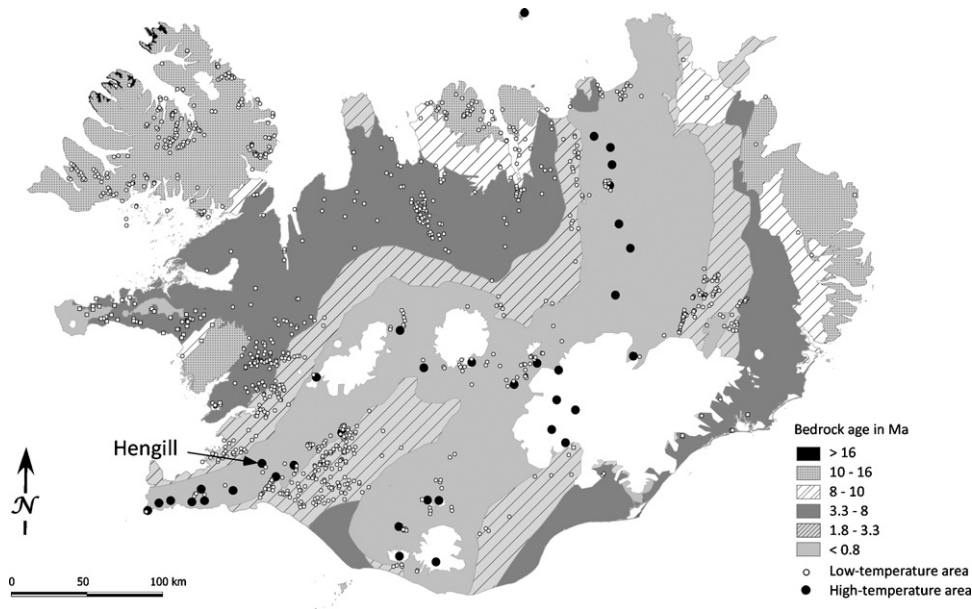


Fig. 1. A simplified geological and geothermal map of Iceland (white areas are glaciers).

junction of the WVZ, the Reykjanes Peninsula (RP), which is the landward extension of the Reykjanes spreading ridge, and the South Iceland Seismic Zone (SISZ), which is a transform zone, transferring part of the crustal spreading from the WVZ to the eastern volcanic zone (Fig. 1). The volcanic products are mainly basalts but with significant occurrences of acidic rocks SW of Mount Hengill.

The Hengill complex hosts three main volcanic centres, which are, listed from SE to NW and decreasing age, Grændalur (0.3–0.5 My), Hrómundartindur (Ölkelduháls) on the decline and Hengill at the peak of activity (Fig. 2). The Grændalur volcano is substantially eroded by glaciation, while Hrómundartindur and Hengill consist mainly of little eroded hyaloclastites. Mount Hengill itself is a tuya (table mountain) with remnants of a lava shield on the top. Both Hrómundartindur and Hengill have erupted in postglacial times. Hrómundartindur last erupted about 11,000 years ago and Hengill 5000 and 2000 years ago (Árnason et al., 1987a).

The Grændalur–Hrómundartindur part of the Hengill volcanic complex is seismically very active. The Hengill central volcano is bisected by a fissure swarm that is 3–5 km wide and about 40 km long, extending N30°E, from the coast to the NE towards Lake Þingvallavatn (Fig. 3). The fissure swarm was active in 1789, with extensive rifting and subsidence (Sæmundsson, 1992). In 1991–2001, high seismic activity occurred in the eastern Hengill area but with different tectonics from that in 1789. Over 90,000 earthquakes were recorded by the permanent seismic network (SIL), operated by the Icelandic Meteorological Office, during that period (Árnason and Magnússon, 2001). Similarly active periods occurred around 1915 and 1950–1955 (Icelandic Meteorological Office, 1950–1955).

Fig. 3 shows that the seismicity is mainly to the east of the fissure swarm. The inferred lineaments show that the tectonics is of a transform character. In June 1998, intense activity started on an ENE–WSW lineament to the SE of Mount Hengill. This was then followed by an earthquake of magnitude >5 and very high activity with epicentres along a SSW–NNE line (the longest solid line in Fig. 3) (Rögnvaldsson, 2000). A GPS geodetic survey had been done just before the episode (Sigmundsson et al., 1997). A repeat survey, right after the episode, showed substantial and consistent crustal movements. The NW and SE quadrants (NW and SE of the intersection of the ENE–WSW and SSW–NNE faults that were active) moved NW and SE, respectively and the NE and SW quadrants moved SW

and NE, respectively. About 5 months later, in November 1998, a relatively big earthquake occurred farther to the south followed by a large swarm of smaller events with epicentres around a WSW–ENE line, as seen in Fig. 3. An aspect of these patterns that is important to this interpretation is that the transform tectonic features terminate in the NW-part of the fissure swarm.

Synthetic Aperture Radar Interferograms (InSAR) show that between 1993 and 1998 there was a considerable uplift going on

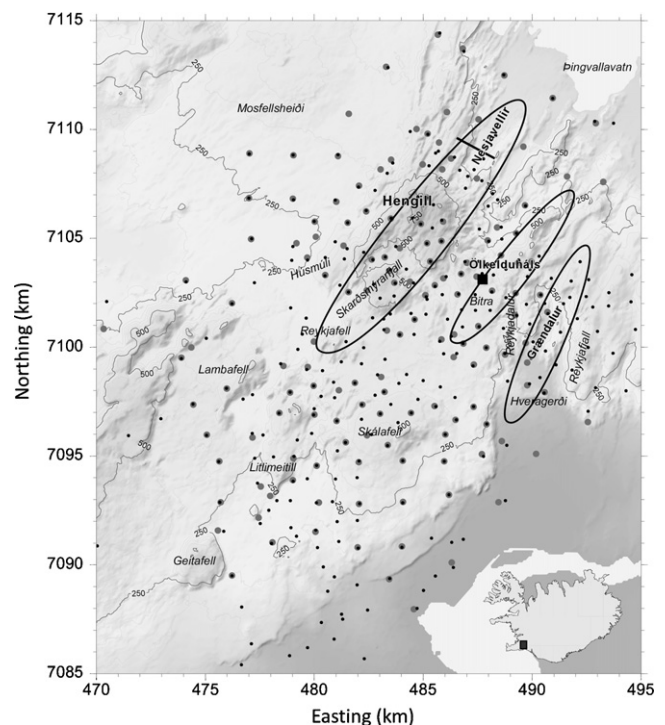


Fig. 2. Location of TEM soundings (small black dots) and MT soundings (big grey dots) considered in this study. Bold black lines roughly outline the three main volcanic centres: Grændalur, Hrómundartindur (Ölkelduháls) and Hengill. The black square shows the location of well ÖJ-1 and the black line at Nesjavellir the location of the profile in Fig. 5. Thin black lines: topographic contour lines in m a.s.l. The insert (bottom right) shows the location of the study area in Iceland.

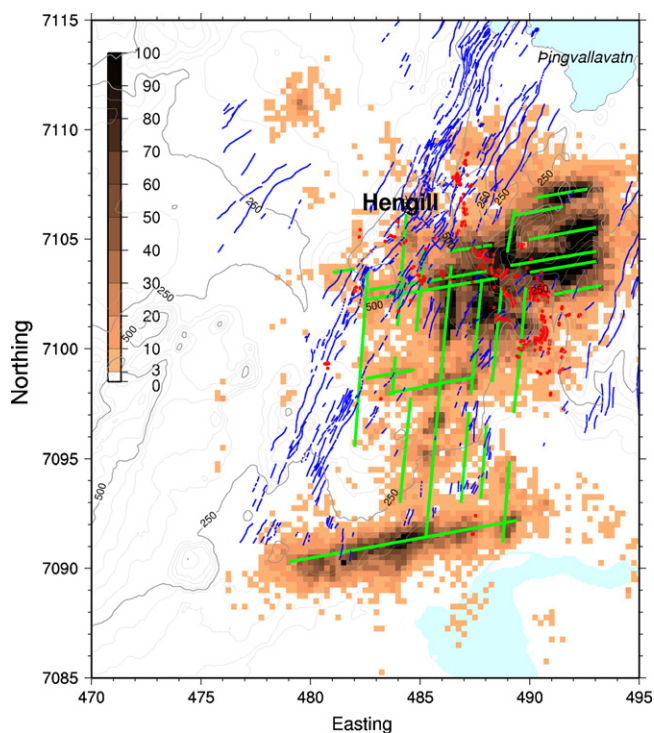


Fig. 3. Density of seismic epicentres (number within 250 m × 250 m bins) from 1991 to 2001 and inferred transform tectonic lineaments (green lines). The lineaments are based on the overall distribution of the seismicity as well as more focused analyses of individual episodes (earthquake swarms). Blue lines: faults and fissures mapped on the surface; red dots: geothermal surface manifestations (Sæmundsson, 1995). Thin black lines: topographic contour lines in m a.s.l. Distances are given in km. Modified from Árnason and Magnússon (2001). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

(up to about 18 mm/year) in the area, with its centre SE of Mount Hengill (Feigl et al., 2000). The InSAR also indicate normal displacement on a fault striking about N5°E (downtrow of about 7 mm to the east) close to the fault that was seismically active in June 1998. The uplift has been interpreted to be due to intrusion of magma into the uppermost crust (Feigl et al., 2000). A simple Mogi (1958) model gives an estimate of the depth to the intrusion of about 7 km, but the uplift could also be caused by an intrusion of larger lateral extent at shallower depth.

3. Brief history of geophysical exploration in the Hengill area

Geothermal exploration in the Hengill area dates back to the 1940s. From 1970 to 1984, Orkustofnun (National Energy Authority of Iceland), the University of Iceland and other universities conducted geophysical surveys in the area, such as DC resistivity surveys using Schlumberger and dipole–dipole soundings (Hersir, 1980; Hersir et al., 1990a), a Bouguer gravity survey (Þorbergsson et al., 1984), an aeromagnetic survey (Björnsson and Hersir, 1981), a seismic refraction survey (Pálmason, 1971) and a passive seismic survey (Foulger, 1984).

In 1984, a new geothermal exploration phase was initiated as part of the project to build a geothermal power plant at Nesjavellir just north of Mount Hengill (Fig. 2). In 1985–1986 a detailed DC resistivity survey (Schlumberger soundings, dipole–dipole soundings, and head-on resistivity profiling) and a Bouguer gravity survey were carried out in the Nesjavellir area (Árnason et al., 1987a). The survey revealed a detailed picture of the resistivity structure in the uppermost one km in that area, and a comparison with borehole data resulted in a new understanding of the resistivity structure of

high-temperature geothermal systems in basaltic rocks discussed below.

The central-loop transient electromagnetic (TEM) method was initially tested in Iceland in 1987. It was found to be superior to DC methods. TEM soundings proved to be much more cost effective and 1D inversion could resolve more complex structures than 1D inversion of DC data. As a result, TEM soundings and later combined use of TEM and MT soundings became the standard electrical method in geothermal exploration in Iceland. Since 1991, about 280 TEM soundings have been carried out in the Hengill area. The size of the source was generally a 300 m × 300 m square loop and with transmitted current around 23 A. The transients were recorded at 30 time gates equally spaced on log scale from 88.13 μs to 68.78 ms after the current turn-off. The locations of the TEM soundings are shown in Fig. 2.

Until about 2004, the main emphasis in geophysical exploration of high-temperature geothermal resources in Iceland was on mapping the resistivity structure and inferring the geothermal alteration mineralogy pattern in the uppermost 0.5–1 km of the crust by TEM soundings. The reason was that mapping the shallow resistivity structure with these soundings had proved to be effective in delineating geothermal reservoirs. In the central-loop TEM method, the coupling to the ground is inductive and no electrodes are installed in the ground. Therefore, soundings can be performed on snow-covered terrain. By using snowmobiles, many places that are inaccessible in the summertime can be reached.

In 2006, a deep well was drilled in the Krafla geothermal area, northern Iceland, into a resistivity feature mapped by TEM soundings that was interpreted as indicating high temperatures. The rocks turned out to be much colder than expected. As a result it became clear that in addition to mapping the shallow resistivity structure a prospecting method was needed to investigate the deeper part of the geothermal system and explore for heat sources. This led to the strategy of using TEM and MT soundings together.

The first MT soundings in the Hengill area were performed in 1976 (Hersir, 1980) and 1982 (Hersir et al., 1990b). After that, they were mainly carried out in four major campaigns, most of them for Reykjavik Energy (OR). The first one was in 2000 when 21 MT soundings were conducted on a 20 km long profile in the SW part of the Hengill area in a test project called Deep Geothermal Prospecting (Oskooi et al., 2005). Between 2000 and 2005 some soundings were made by ÍSOR. The second campaign was in 2005 when 59 MT soundings were recorded, in the so-called INTAS project, supported by the European Union. The third campaign was in 2006 within the I-GET project when a total of 70 soundings were made. Finally, ÍSOR has added 30 more soundings. Care was taken to make the measurements at the same (or nearly the same) places where TEM soundings had been done earlier, but that was not possible for all of them.

Due to poor data quality and reformatting problems of the older MT data, only those from the last three campaigns are included in the present study; a total of 159 soundings. Eleven were excluded because of their poor quality, so the results presented here are based on 148 MT soundings. Their location is shown in Fig. 2. All these soundings were acquired using Phoenix MTU instruments with a remote reference station. In the 2005 and 2006 campaigns three five-component and two two-component units were used. The deployment was planned such that the two-component stations were located close to a five-component station. The stations were moved every 24 h so the effective recording time was about 20 h per station. The remote reference station was about 10 km from the nearest survey station.

Three major parallel power lines that are close to each other cross the survey area. This caused considerable noise in the soundings near (about 100–200 m) the power lines. The low-frequency signal was weak because of low solar activity. The shortest period

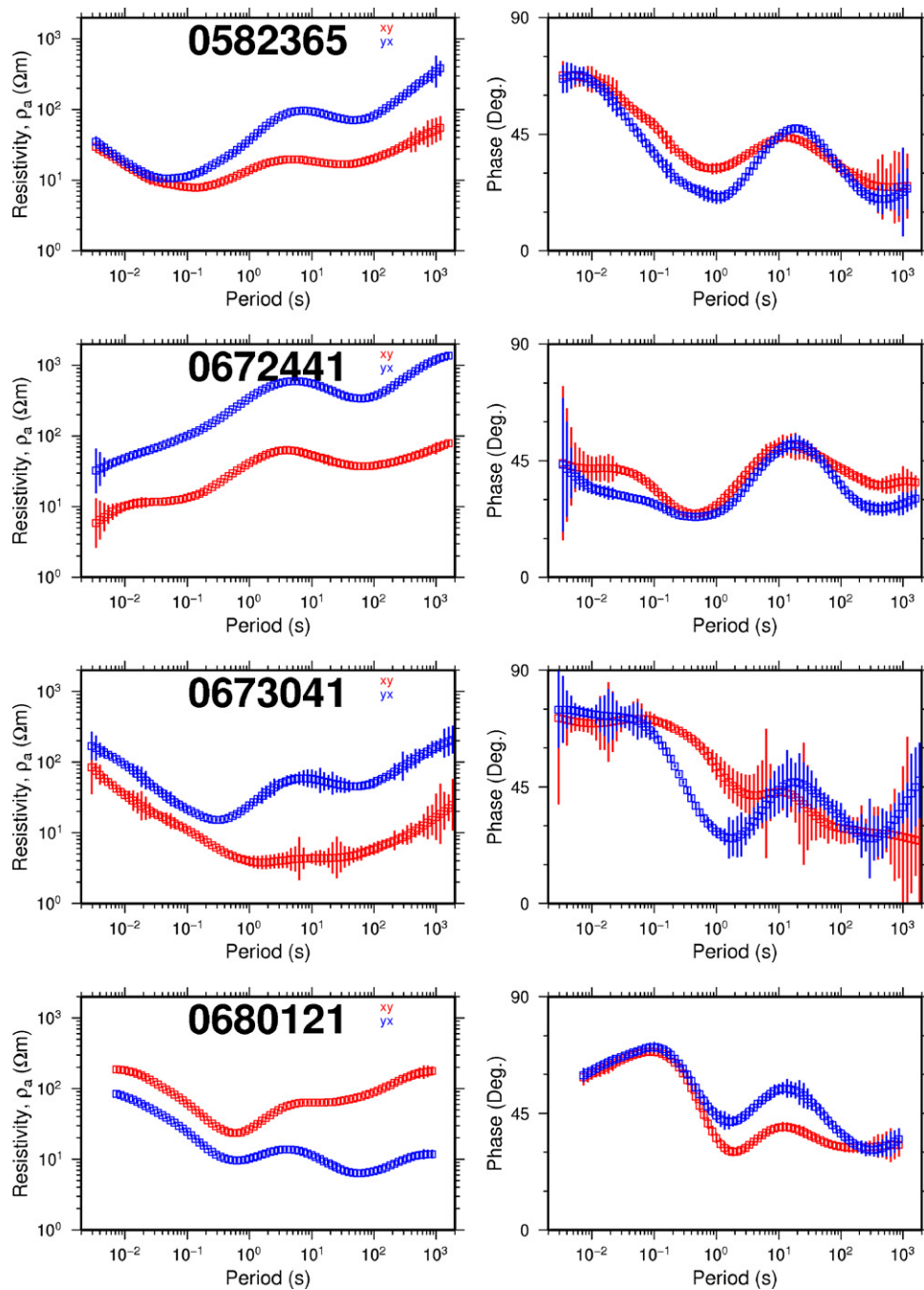


Fig. 4. Four typical examples of apparent resistivity and phase curves of MT soundings (in the measured magnetic north direction) in the Hengill area (red and blue symbols are xy and yx data, respectively). The figure shows examples of severe static shifts, where parallel red and blue curves are widely separated at periods under 1 s. A non-1D resistivity pattern is indicated by the non-parallel divergence of the red and blue apparent resistivity curves for periods longer than 1 s. The bold numbers in the resistivity panels correspond to station numbers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

recorded was 0.0034 s and the longest period in the 100–1000 s range. Fig. 4 shows four typical examples of apparent resistivity and phase curves of MT soundings in the Hengill area.

4. The resistivity structure of high-temperature geothermal systems in basaltic rocks

Studies of fresh water, high-temperature geothermal systems in Iceland show that they share a common resistivity structure (Árnason et al., 2000) characterised by a low-resistivity cap (1–10 Ωm) on the top and outer margins of a more resistive core deeper in the reservoir. That core has about an order of magni-

tude higher resistivity (Árnason et al., 1987b; Árnason and Flóvenz, 1992; Eysteinnsson et al., 1994; Árnason et al., 2000; Kristinsdóttir et al., this issue). Initially, this was puzzling because it was believed that the resistivity should decrease with increasing temperature. Because the solution to the “puzzle” in Iceland was found in the Hengill area, it is briefly summarised here.

As mentioned above, a detailed multi-method DC resistivity survey was conducted in the Nesjavellir area in 1985–1986. The joint 2D interpretation of the different datasets gave a detailed picture of the resistivity structure in the uppermost one km, which could be compared with the lithology, alteration mineralogy and temperature distribution encountered in nearby boreholes. The resistivity

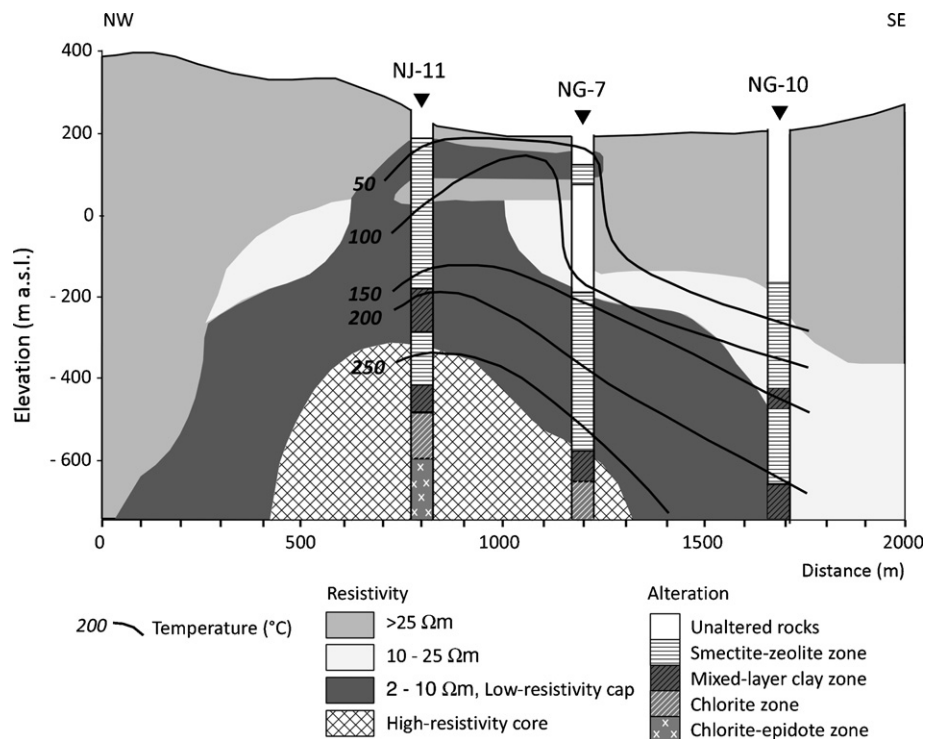


Fig. 5. Resistivity cross-section for the Nesjavellir field constructed on the basis of 2D joint interpretation of DC data, alteration mineralogy data from wells NJ-11, NG-7 and NG-10, and interpolated temperatures from these and other nearby wells (Árnason et al., 1987a). See Fig. 2 for location.

was not found to correlate with lithology, but a fairly clear and consistent correlation was found with alteration mineralogy.

Fig. 5 shows a simplified resistivity cross-section from Nesjavellir, the dominant alteration mineral zones in three wells, and interpolated temperatures in these and other nearby wells. The resistivity is high in the surrounding cold, unaltered rocks. At 50–100 °C, pronounced geothermal alteration takes place with smectite and zeolites as the dominant alteration minerals, making the rocks conductive. At higher temperature, in the 220–240 °C range, the smectite and zeolites are gradually replaced by chlorite as the dominant alteration mineral in the mixed-layered clay zone (Kristmannsdóttir, 1979) and the resistivity increases again. At still higher temperatures epidote becomes abundant. The smectite and the zeolites have loosely bound cations that make these minerals conductive, whereas in the chlorites all ions are bound in a crystal lattice (Deer et al., 1962), and are, therefore more resistive.

A similar correlation between resistivity and alteration mineralogy has been found in all the explored and drilled high-temperature geothermal fields in Iceland, as well as in many high-temperature areas in other countries (Uchida, 1995; Pellerin et al., 1996; Árnason et al., 2000; Ussher et al., 2000). This was an important finding, because if the temperatures that produced the alteration mineralogy prevail, the resistivity structure can be used to predict subsurface temperatures. Since generally the alteration is unaffected by subsequent cooling, in most cases the resistivity structure can, therefore, be regarded as a “maximum thermometer”. However, alteration minerals sometimes have indicated a lower temperature than measured in the wells. This was interpreted as indicating a young system still heating up with alteration lagging behind and, therefore, not in thermal equilibrium.

There are several examples where drilling has shown that the reservoir (or parts of it) has cooled down after high-temperature alteration developed. To the SE of Hengill, a resistive core, indicative of temperature above 240 °C in an undisturbed thermal regime, was found as shallow as at 200 m depth. Well ÖJ-1 (Fig. 2) encountered temperatures lower than 200 °C to a depth of 1000 m (Árnason and

Magnússon, 2001). However, the alteration mineralogy agreed well with the resistivity structure revealed by a nearby TEM sounding (Fig. 6). Some parts of the Krafla geothermal reservoir show similar cooling in the uppermost one km but with alteration in agreement with the resistivity (Árnason et al., 2000). A well drilled in an area of Krafla where TEM resistivity pattern was interpreted as indicating “high-temperature” turned out to be surprisingly cold, below 100 °C between 700 and 1400 m depth, and gradually increasing to 240 °C at the bottom of the well at 2900 m (Mortensen et al., 2009). The alteration was, however, in good agreement with the resistivity.

These examples demonstrated that, in order to avoid unproductive investments in drilling, it is not enough to map the shallow alteration, but information about the presence (or absence) of a deep heat source is necessary. Results of a subsequent MT survey conducted in Krafla (Árnason et al., 2009) exhibited no sign of deep heat sources under the area where the unproductive well was drilled, but they were found below the productive region of the field.

5. 1D inversion of TEM soundings

All the TEM soundings in the Hengill area have been interpreted using 1D (layered earth) Occam (minimum structure) inversion (Constable et al., 1987). Fig. 7 shows a typical TEM apparent resistivity curve and its interpretation. The 1D models of the individual soundings were compiled and gridded to make a 3D resistivity model of the uppermost part of the crust.

Resistivity maps for the Hengill area at sea level and at 500 m below sea level (m b.s.l.) are presented in Fig. 8(a) and (b), respectively. Few of the TEM stations located at higher elevations do not detect the deeper resistivity because it is below the depth of exploration of the soundings. Therefore, the 500 m b.s.l. map has limited resolution in higher elevation areas, but in lower regions, the map indicates that the resistive core extends along the fissure swarm to the SW. In these figures, the areas through the resistive core are

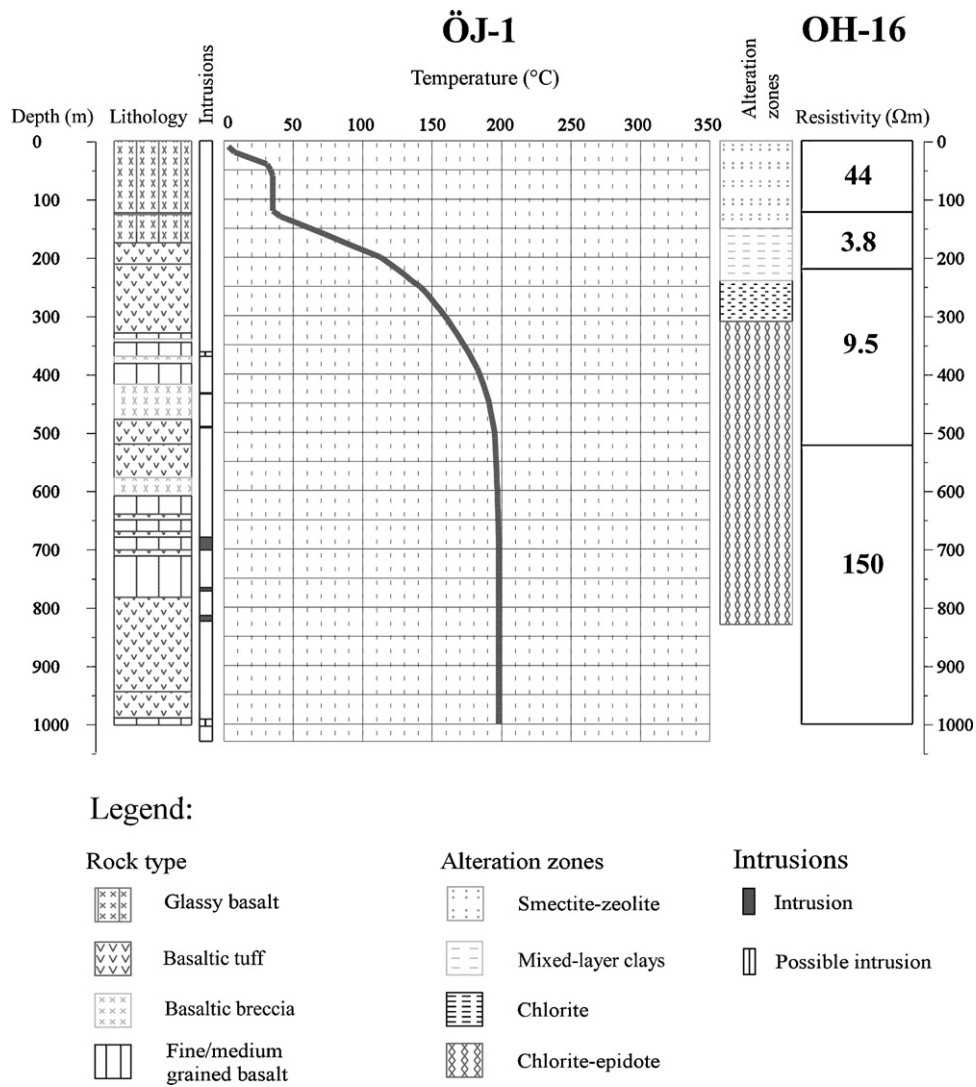


Fig. 6. Lithology, temperature and alteration mineralogy in well ÖJ-1 (location shown in Fig. 2) and interpreted resistivity layering from the nearby TEM sounding OH-16 (modified from Árnason and Magnússon, 2001).

shown as red hatched. To highlight the interpretation elements, the plotting program assumes very low resistivity (1 Ω m) just outside the resistive core to let the contouring routine outline the core.

At sea level (Fig. 8(a)), a high-resistivity core zone is found beneath and to the SE of Mount Hengill (at 400–800 m depth); high-temperature mineral alteration is also seen at this depth. But as stated above, drilling has shown that the rocks to the SE of Hengill have undergone cooling at least in the uppermost one km. At greater depth a continuous resistive zone enclosed by lower resistivity rocks stretches to the SE and NW of Mount Hengill (i.e. in the crustal spreading direction). Such resistive zones seen to the SSW, within the fissure swarm, appear to be discontinuous because the top of the high-resistivity zone is below the depth of exploration of the TEM soundings.

6. The MT static shift problem

The MT method, and all resistivity methods that are based on measuring the electric field at ground surface, suffer from the static shift problem. This phenomenon is caused by resistivity inhomogeneity close to the electric dipoles. There are mainly two phenomena that produce static shifts, i.e. voltage distortion (dependence of the electric field on the resistivity where the voltage is measured) and current distortion (current channelling). Rough

topography can also produce static distortion because near-surface currents are gathered in valleys and current density is reduced on ridges and mountains.

The problem is that the electric field on the surface and consequently the apparent resistivity is scaled by an unknown dimensionless factor (shifted on log scale) due to resistivity heterogeneity in the vicinity of the measuring dipole. This shift is independent of frequency and persists at all frequencies. Static shift can be a big problem in volcanic geothermal areas where resistivity variations are often extreme. Shift factors, S , of the apparent resistivity have been observed as low as 0.1 and as high as 3 (Árnason, 2008). Sternberg et al. (1988) have published results of model calculations showing that the voltage distortion and current channelling can produce dramatic static shifts in MT data. A shift down by $S=0.1$ will result in 10 times too low resistivity values and about three times too small depths to resistivity boundaries (Árnason, 2008). The problem is made even worse by the fact that often the shifts are not random. Sometimes all soundings in large areas are shifted upwards or all are shifted downwards. An extreme example of the problem was found in a recent combined TEM/MT survey conducted in Djibouti in the Asal Rift, East Africa (Árnason, 2008).

The use of the MT data themselves has been proposed to identify and correct for static shifts. This is commonly attempted in 2D inversions of MT soundings made along profiles. Static shift cor-

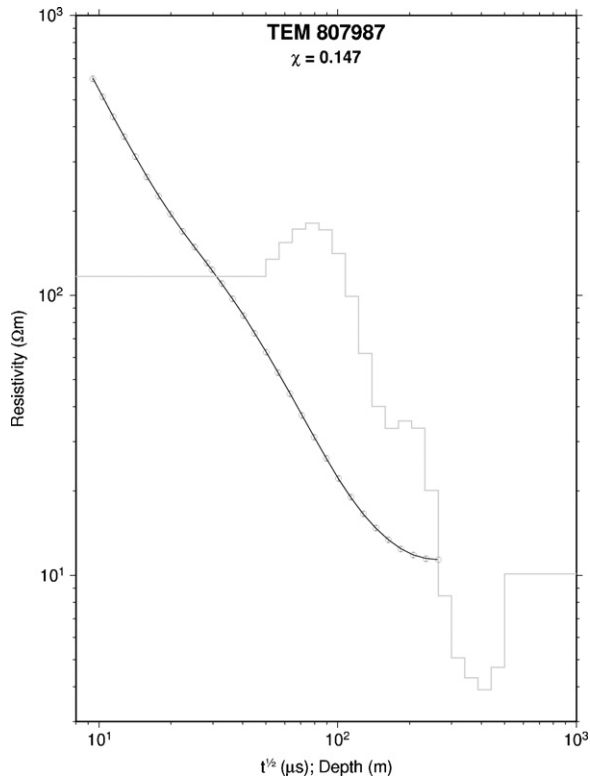


Fig. 7. A typical TEM sounding from the Hengill area and its 1D inversion. Grey circles: measured late-time apparent resistivities; black line: apparent resistivity calculated from the model shown in grey. The number on the top of the figure (TEM 807987) corresponds to the name of the station.

rections based on the MT data usually employ spatial averaging and/or statistical assumptions about the shifts (see e.g. deGroot-Hedlin, 1991; Ogawa and Uchida, 1996 and references therein). The approach by deGroot-Hedlin (1991) assumes that the shift

multipliers are random and that their product is close to one for a sufficiently large number of soundings. It will be seen in the next section that this assumption is not justified. One approach to separate and correct for distortion of the impedance tensor is the Groom–Bailey tensor decomposition (Groom and Bailey, 1989), but in that approach a scaling factor is still undetermined.

In the central-loop TEM method, the measured signal is the decay rate of the magnetic field (induction in a coil) from the current distribution induced by the current turn-off in the source loop. As these currents diffuse outwards and downwards with time, they are only dependent on the near-surface resistivity structure at very early times. At late times the distortions due to near-surface inhomogeneity disappear. This has been thoroughly tested by model calculations (Sternberg et al., 1988). It can also be argued that current distortion at shallow depth due to topographic effects has vanishing influence on the TEM data at late times when the induced currents have diffused deep below the surface. Therefore, central-loop TEM soundings can be used to correct for static shifts in MT soundings (Pellerin and Hohmann, 1990).

7. Joint 1D inversion of TEM and MT soundings

Because of the static shift problems in MT data in volcanic geothermal fields, in the Hengill area care was taken to locate TEM soundings at the same places (within about 200–300 m) as the MT soundings. Software was developed within the I-GET project for joint 1D (layered earth) inversion of central-loop TEM soundings and MT soundings (Árnason, 2006); it offers several options for inversion of TEM and MT data separately, or jointly. In the joint inversion, one more parameter is inverted, in addition to the layered model resistivity and thickness parameters, namely a static shift multiplier by which the MT apparent resistivity has to be divided so that both the TEM and MT data can be fitted with the same model. The program can do both standard layered inversion (inverting resistivity values and layer thicknesses) and Occam (minimum structure) inversion with exponentially increasing layer thicknesses with depth. It offers a user-specified damping of first-

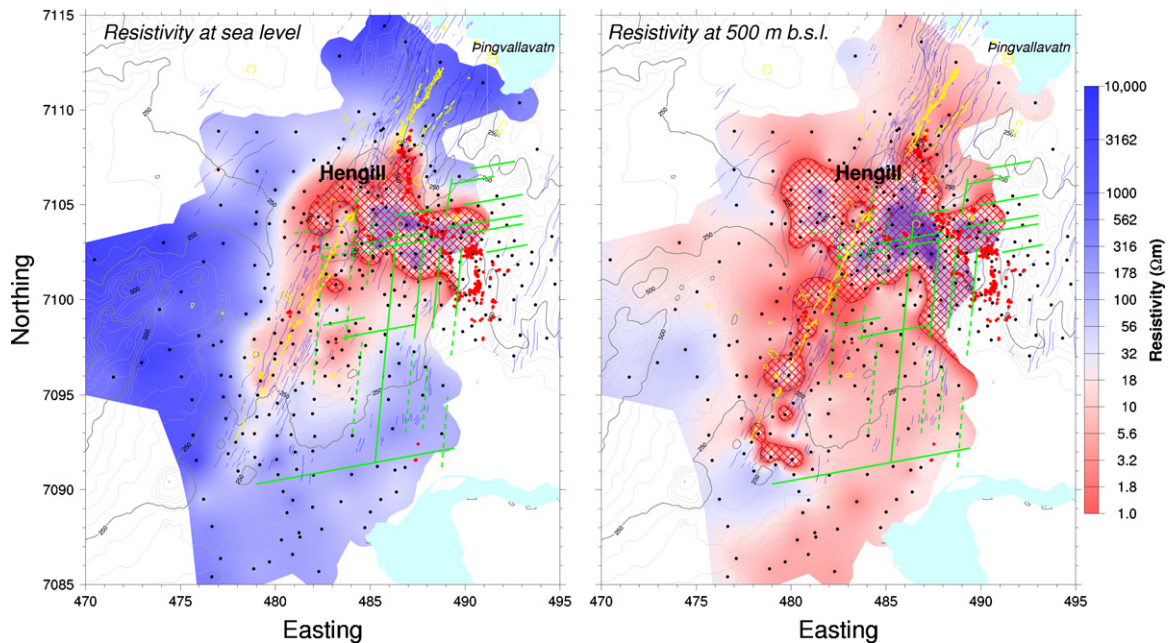


Fig. 8. Resistivity in the Hengill area based on 1D inversion of TEM data: (a) at sea level elevation; (b) at 500 m b.s.l. Faults and fractures mapped on the surface are shown in blue, faults inferred from seismicity in green and volcanic craters and fissures in yellow. Red dots: geothermal surface manifestations; black dots: TEM stations; red-hatched areas: resistive core; thin black lines: topographic contour lines in m a.s.l. Distances are given in km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

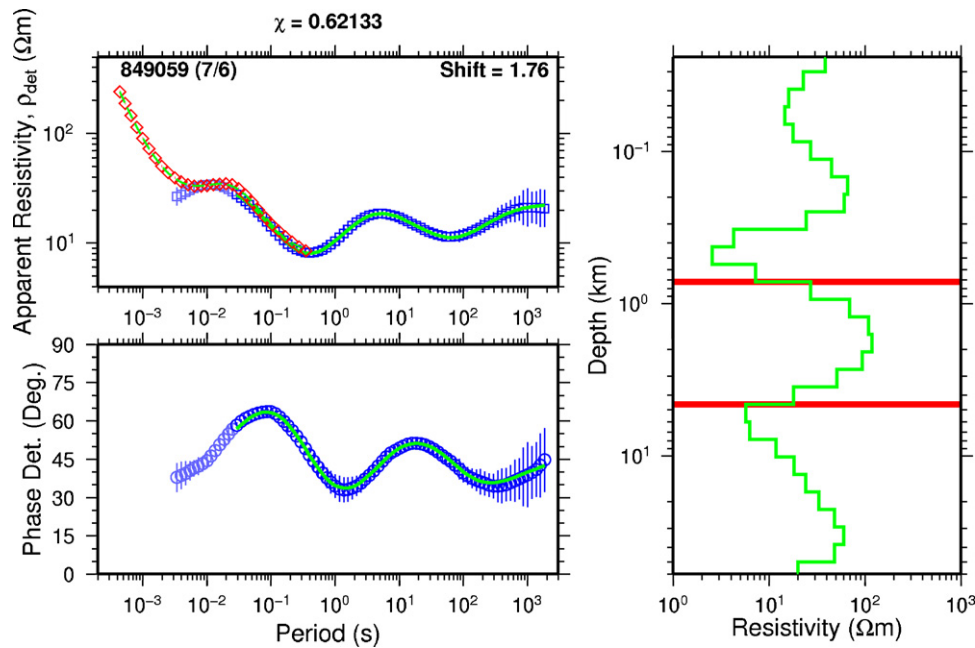


Fig. 9. Typical result of a joint 1D inversion of TEM and MT soundings. Red diamonds: TEM apparent resistivities transformed to a pseudo-MT curve; blue squares: measured apparent resistivities; blue circles: apparent phase derived from the determinant of MT impedance tensor; light blue symbols to the left of the green curve: data not used in the inversion; green lines; vertical blue lines: error bars. Right panel: results of the 1D resistivity inversion model, in the left panels its synthetic MT apparent resistivity and phase response. Red horizontal lines on the right panel: transitions mapped as elevations in Fig. 14. Number on the top of the figure (0581435): name of the MT station; number in the upper left panel (849059): name of the TEM station. The numbers in parenthesis (7/6) indicates that the two stations were 7 m apart and their elevation difference was 6 m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

order (sharp steps) and second-order derivatives (oscillations) of model parameters (logarithm of resistivity and layer thicknesses).

A joint 1D Occam inversion has been performed for the rotationally invariant determinant apparent resistivity and phase of the Hengill soundings and associated TEM soundings. Fig. 9 shows a typical result of such an inversion. The best estimate of the shift parameter for this MT sounding is $S = 1.76$ (upper right hand corner of the apparent resistivity panel), i.e. the MT apparent resistivity has to be divided by 1.76 to be consistent with the TEM sounding. In Fig. 9 the TEM data are plotted as a function of the period (T) by using the transformation $T = t/0.2$ (Sternberg et al., 1988) and not as a function of square root of time (t) as in Fig. 7.

A histogram of the shift multipliers for all the 148 Hengill MT soundings is given in Fig. 10. It is seen that they are in the 0.2–2.2 range, and that the shift multipliers lower than one are more common than those higher than one. Table 1 presents some statistics on the shifts in Hengill, as well as in the Krafla and Asal Rift high-temperature geothermal areas (Árnason, 2008). For Hengill, the mean of the shift multipliers is about 0.9 and the geometric mean, about 0.86. The product of all the shifts values is very low, i.e. in the order of 10^{-10} . The table shows similar statistics for the Krafla field but the means and the product are considerably lower in the Asal Rift.

It is evident from the data in Table 1 that the assumption of deGroot-Hedlin (1991), that the product of the shift multipliers for a large number of MT soundings should be close to one, is in general not justified. Fig. 11 shows the spatial distribution of the shift multi-

pliers in the Hengill area, indicating that there are some areas where the MT apparent resistivity is consistently shifted downwards and others where it is shifted upwards.

Fig. 12 shows maps of resistivity in the Hengill area at 0.8, 1.2, 4.5 and 6.5 km b.s.l., according to the joint 1D inversion of TEM and determinant MT data. Here the resistivity is drawn directly

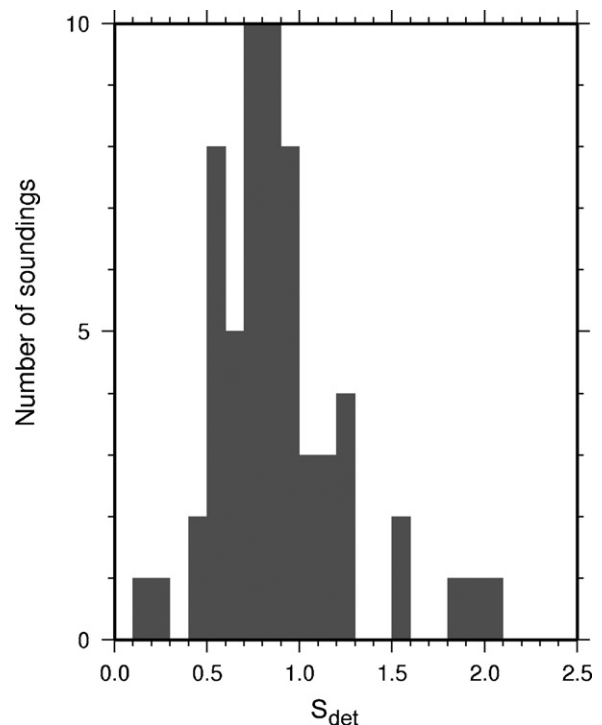


Fig. 10. Histogram of static shift parameters for determinant apparent resistivity in the Hengill area.

Table 1
Statistics of shift multipliers in the Hengill, Krafla and Asal Rift (Djibouti) areas.

Area	Number of soundings	Product	Arithmetic mean	Geometric mean
Hengill	149	1.089×10^{-10}	0.905	0.855
Krafla	124	1.683×10^{-10}	0.969	0.834
Asal Rift	64	1.045×10^{-16}	0.702	0.563

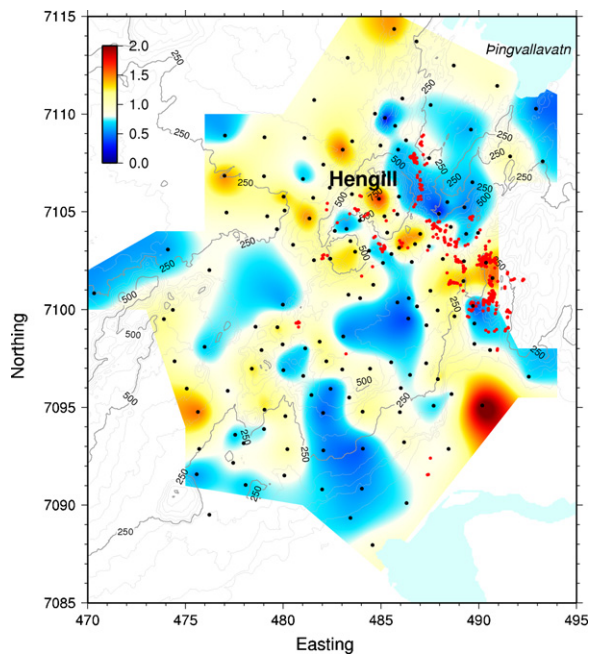


Fig. 11. Spatial distribution of static shift parameters for determinant apparent resistivity in the Hengill area. Red dots: geothermal surface manifestations; black dots: MT soundings; thin black lines: topographic contour lines in m a.s.l. Distances are given in km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

from the models and the resistive core is not emphasized by red-hatched shading and very low resistivity at its boundaries as in Fig. 8. The low-resistivity layer reflecting smectite-zeolite alteration is encountered at shallow depth, but at 1.2 km b.s.l. the resistive zone that is interpreted as indicating high-temperature alteration extends almost over all the survey area.

At about 3 km b.s.l. a deep low-resistivity zone starts to appear under Mount Hengill, and Fig. 12 shows that at 4.5 km b.s.l. the low resistivity has spread out and a deep conductor has appeared farther to the SW. At 6.5 km b.s.l. a widespread deep conductor is observed. At about 10–15 km b.s.l. the resistivity increases again, reaching about 100 Ω m at greater depths. In Fig. 13 a W–E resistivity cross-section for the Hengill area is shown for two different depths.

Almost all the MT soundings in the Hengill area and the resulting models have the same character as the ones shown in Figs. 4 and 9, i.e. two low-resistivity layers. The nature of the uppermost low-resistivity layer and the increasing resistivity below is due to hydrothermal mineral alteration as discussed earlier. The nature of the deep conductive layer is not well understood; possible explanations will be discussed later.

The elevation of the lower boundary defined by the steepest resistivity gradient (Fig. 9) of the uppermost conductive layer (corresponding to the top of the resistive core zone) and the top of the deeper low-resistivity layer are mapped in Fig. 14. The figure shows that the resistive core zone domes up under and to the ESE of Mount Hengill, extending at relatively shallow depth to the west of Hengill and to the SSW along the fissure swarm, but dips towards the NE and SW.

Fig. 14 indicates that the deep conductor is shallowest under a WNW–ESE elongated area centred under Hengill, similar to that of the resistive core. In addition, the figure shows conductors at relatively shallow depth, at about 4 km b.s.l., on both sides of the fissure swarm around 10 km south of the Hengill. The top of the resistive core zone, to some extent, appears to be controlled by the fissure swarm, but the main anomalies at depth, i.e. the resistive

core and the deep conductor, have a WNW–ESE elongation through Hengill. Fig. 14 indicates that this direction is the same as that of a zone where E–W faults, inferred from seismicity, meet N–S faults associated with detected seismic events. The region where the top of the resistive core is highest coincides with the area where the seismicity was most intense in 1991–2001, and also with the centre of uplift according to InSAR (Feigl et al., 2000).

8. 3D inversion of MT soundings

Until recently only 1D and 2D inversions have been available for routine interpretation of MT data. Five-component MT soundings are tensor data and individual soundings contain information on both vertical and lateral resistivity variations (e.g. resistivity strike). In contrast, TEM (vertical component) soundings are scalar data and individual soundings give no information on lateral variations, although these can be inferred by considering soundings at different locations.

In a 1D inversion it is assumed that resistivity only varies in the vertical direction. Due to the tensor nature of the MT data there are several options for doing 1D inversion, i.e. inverting (a) the apparent resistivity and phase derived from either the off-diagonal elements of the impedance tensor (xy or yx modes); (b) the average of the off-diagonal elements of the tensor; or (c) the determinant, which is rotationally invariant. In cases of complicated resistivity structure, these various alternatives can give very different results. Choosing the electrical strike direction for rotation and 1D inversion of either the xy or yx modes, or choosing another average method is often somewhat subjective and unreliable. Preliminary model calculations and reported geothermal industry experience indicate that smooth, minimum structure 1D inversion of the rotationally invariant determinant data can reproduce complex 3D resistivity structures, while 1D inversion of either xy or yx modes can give misleading results.

In a 2D inversion it is assumed that the resistivity can vary with depth and in one lateral direction, and that the resistivity is constant in the other horizontal direction (electrical strike). The inversion is done for soundings on profiles which are roughly perpendicular to the electric strike. If the field setup coordinate system is not parallel and perpendicular to the electrical strike, the MT impedance tensor data are mathematically rotated with one axis perpendicular to the electrical strike and the other axis parallel to it. The inversion is then done to fit the apparent resistivity and phase data computed from the impedance elements with the electric field perpendicular to the strike (called the TM-mode) or the mode with the field parallel to the strike (called the TE-mode), or both simultaneously. In this study, 2D inversion was not applied to the Hengill data because the dimensional indicators of the impedance tensor indicated that the resistivity structure was 3D at periods longer than 0.5–1 s.

As computing power has increased, codes have been developed for 3D inversion of MT data (e.g. Mackie and Madden, 1993; Newman and Alumbaugh, 2000; Zhdanov et al., 2000; Farquharson et al., 2002; Newman et al., 2008). The 3D MT inverse problem is an under-determined (ill-posed) problem because the number of unknown parameters in the discretized model is normally much higher than the number of measured values. The instability of these inversions is mitigated by adding smoothness constraints on the model parameters. However, the number of model parameters makes inversion computationally intensive.

To reduce the computational cost, in this study we used the 3D MT inversion code WSINV3DMT (Siripunvaraporn et al., 2005), which utilizes a formulation of the inverse problem where the smooth inversion is done in the data-space rather than in the model-space (Siripunvaraporn and Egbert, 2000). This reduces the dimensionality of the problem dramatically and makes it possi-

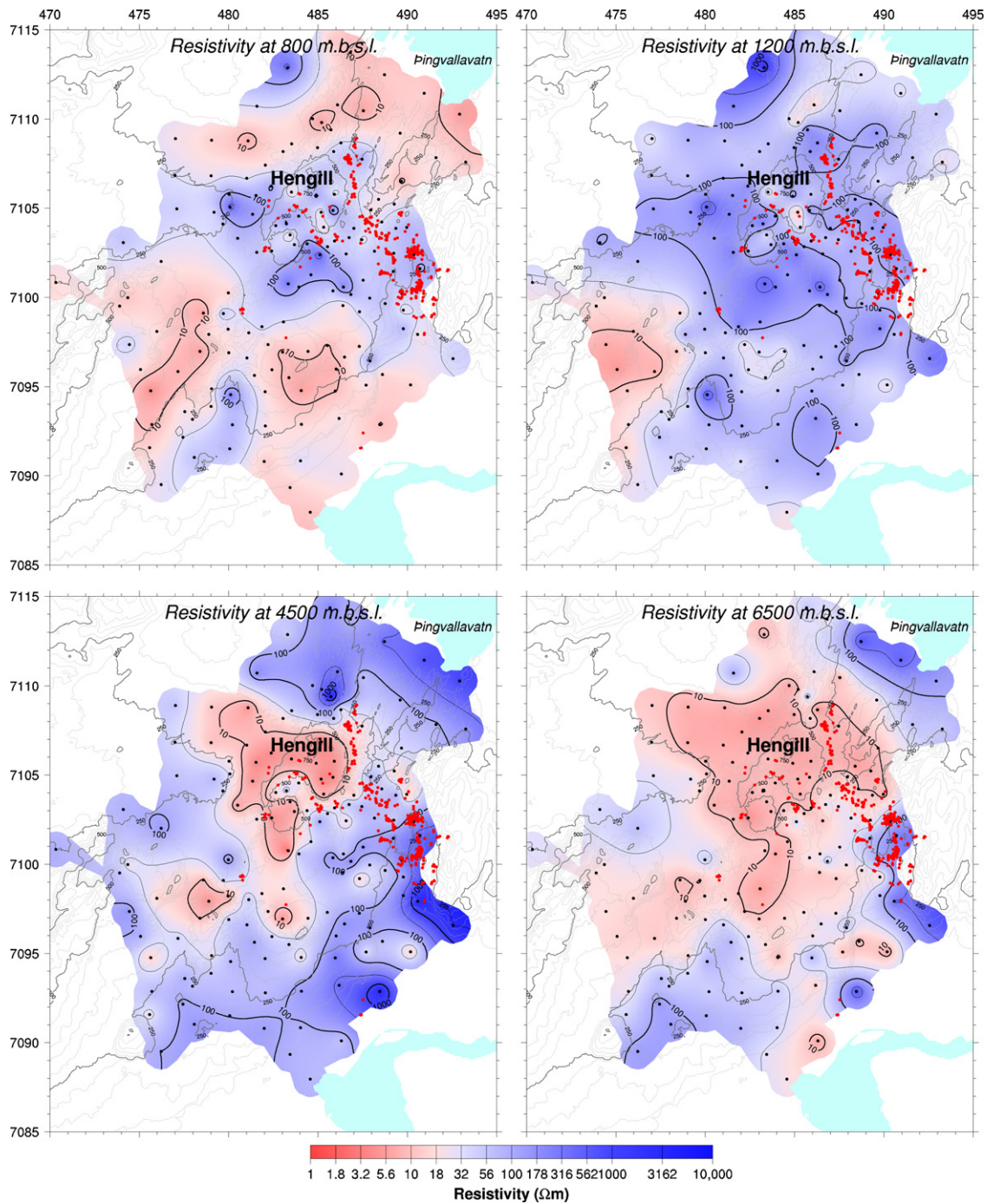


Fig. 12. Resistivity in the Hengill area at 0.8, 1.2, 4.5 and 6.5 km b.s.l. according to joint 1D inversions of TEM and determinant MT data. Red dots: geothermal surface manifestations; black dots: MT soundings; thick black lines: resistivity contour lines; thin black lines: topographic contour lines in m a.s.l. Distances are given in km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

ble to run inversion on a single-processor UNIX/Linux workstation. The forward algorithm is a staggered grid finite-difference solution of the Maxwell's equations. The code inverts the impedance tensor, either the full tensor or its off-diagonal elements. The 3D code used here assumes flat surface. This is a simplification that has the consequences that resistivity values in the model refer to depth below the surface rather than elevations. This has to be kept in mind when modelling shallow levels, but at greater (>1 km) depths the effect of this simplification is assumed to be negligible. Assuming a flat surface also prohibits simulating current distortion due to topography, but, as stated before, using TEM data for static

shift correction will also correct, at least to some extent, for this distortion.

The first step in the 3D inversion was to set up a $40 \times 40 \times 25$ element mesh for the model. Due to the expected influence of the fissure swarm and other geological structures, it was decided to have the x-axis parallel to the geological strike, i.e. N30°E, which is the general strike direction in the MT data. The grid has 40 vertical grid planes at constant x values and 40 at constant y values with 1 km spacing to ± 10.5 km from the centre and then progressively increasing in size to ± 110.5 km. The number of horizontal grid planes is 25, where the first plane is the ground surface (consid-

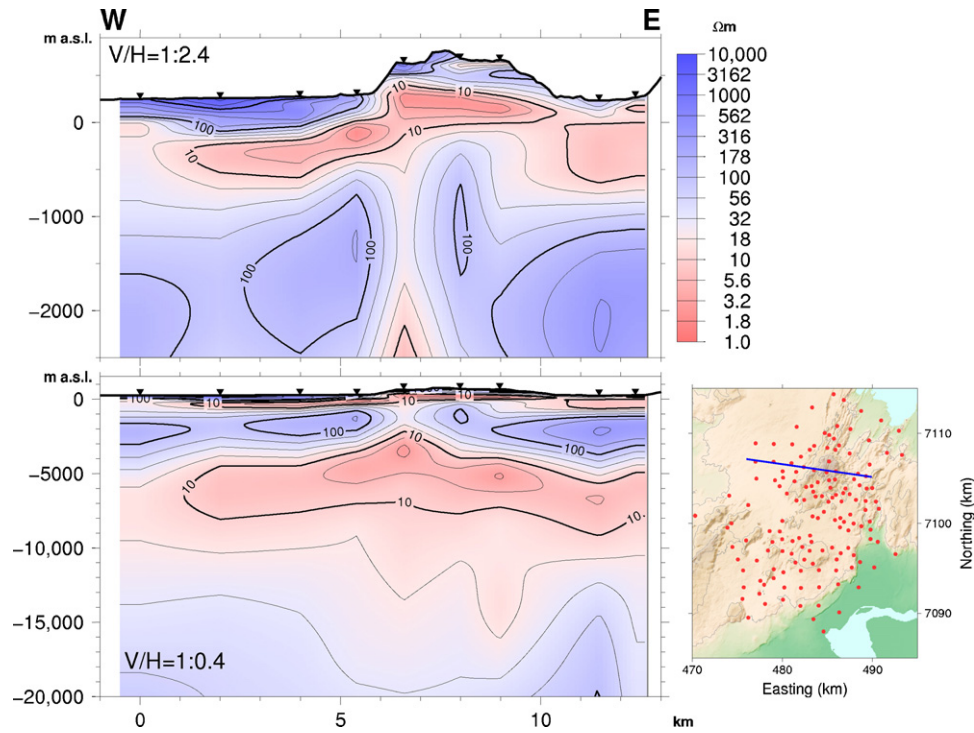


Fig. 13. West-east resistivity sections across the Hengill area for two different depth ranges obtained from stitched joint 1D inversions of TEM and determinant MT data. Inverted triangles: MT stations; V/H: ratio between vertical and horizontal axes. Section location is shown as a blue line in the map to the right. Red dots in that map: MT stations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

ered to be flat), the second plane is at 0.1 km below the surface, the next five are at a 0.2 km interval (5×0.2 km) down to 1.1 km. The following four planes are at a 0.5 km interval (4×0.5 km) to 3.1 km depth. The next planes are at 3.9, 4.9, 5.9, 7.9 and 9.9 km depth and subsequent grid planes are separated at gradually increasing spacing down to 91 km depth.

Fig. 15 shows the location of vertical grid planes in the central part of the 3D model grid. The MT soundings are shown as points; the circles show the location of the subset of 60 soundings chosen for inversion. The inversion algorithm requires that the soundings are at the centre of the grid cells and the data from the soundings nearest to the circles are inverted. The impedance tensors for the

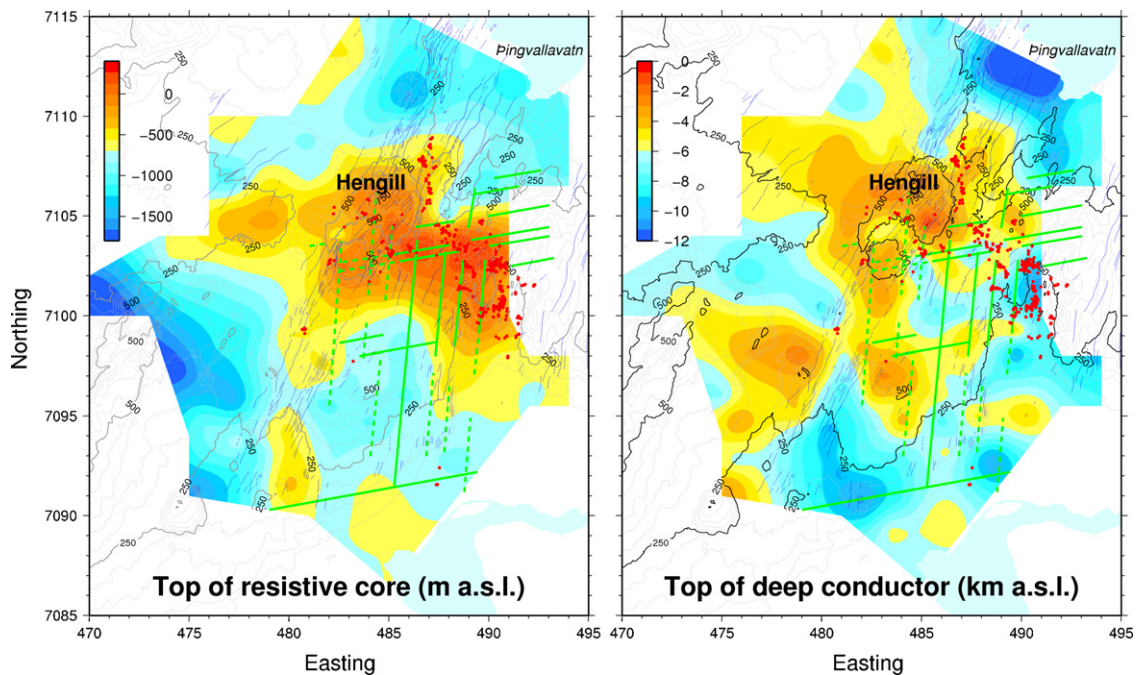


Fig. 14. Elevation of the resistive core (left) and of the top of the deep conductor (right) in the Hengill area, based on joint 1D inversions of TEM and determinant MT data. Red dots: geothermal surface manifestations; blue lines: faults and fractures mapped on the surface; green lines: faults inferred based on seismic data; thin black lines: topographic contour lines in m a.s.l. Distances are given in km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

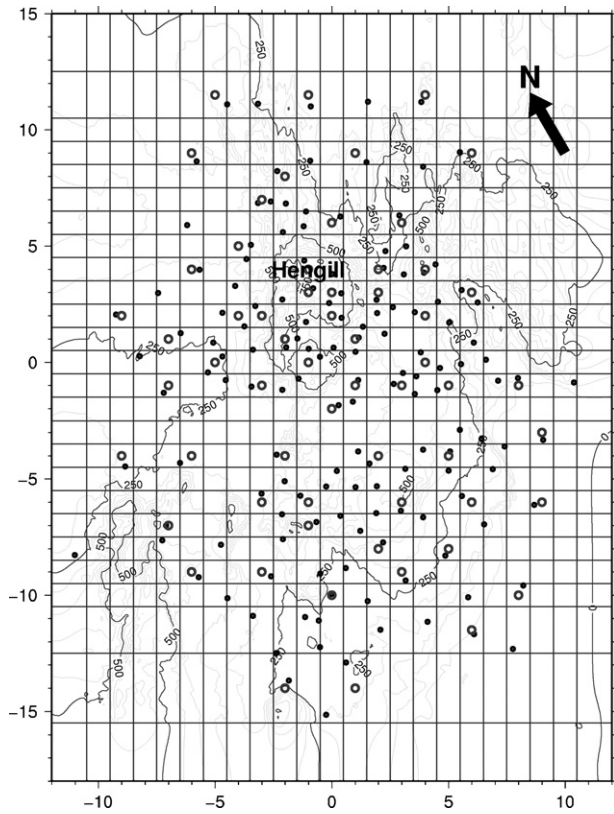


Fig. 15. Horizontal slice showing the vertical grid planes in the 3D grid (grid coordinates are in km). Black points: MT soundings; open circles: location of soundings chosen for inversion. Data from soundings nearest to the open circles were inverted; black lines: topographic contour lines in m a.s.l.

chosen soundings were rotated from the field setup (i.e. the x -axis parallel to the magnetic north, $N16^\circ W$) to the geological strike of $N30^\circ E$.

9. Static shift correction of the MT tensor

Since the inversion is performed on the MT impedance tensor, a static shift correction of its elements is needed. In the static shift correction it is assumed that the shifts are mainly due to distortion

of the electric fields. If S_{xy} and S_{yx} are the shift multipliers for the xy and yx modes, respectively, the corrected tensor $\mathbf{Z}^c$ is given by

$$\begin{bmatrix} Z_{xx}^c & Z_{xy}^c \\ Z_{yx}^c & Z_{yy}^c \end{bmatrix} = \begin{bmatrix} C_x & 0 \\ 0 & C_y \end{bmatrix} \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix}; \quad C_x = \sqrt{\frac{1}{S_{xy}}}; \quad C_y = \sqrt{\frac{1}{S_{yx}}}$$

This formulation is a simplification of the total static distortion discussed by Groom and Bailey (1989), where twist and shear are not considered. It is similar to the anisotropy part in the Groom–Bailey decomposition that has no control of the absolute values of C_x and C_y , only of their ratio. The shift multipliers S_{xy} and S_{yx} were determined by separate joint inversion of the xy and yx modes of the subset of 60 soundings chosen for the inversion and the associated TEM sounding. An example is shown in Fig. 16, where $S_{xy} = 1.10$ and $S_{yx} = 0.68$.

A histogram of shift multipliers for the soundings used in the 3D inversion is presented in Fig. 17. All the multipliers, both S_{xy} and S_{yx} , are shown on the left, and their ratio, S_{xy}/S_{yx} , on the right. The distribution of shift multipliers in this figure is similar to that of Fig. 10, but both lower and higher values than for the determinant data are observed. This is not surprising since the determinant is a type of average of the impedance tensor. If one of the xy or yx polarizations is shifted up and the other down, the determinant apparent resistivity takes a sort of “average” of the shifts. The right part of Fig. 17 indicates that shift multipliers S_{xy} and S_{yx} are frequently very different for a given MT sounding that has its x -axis rotated to align with the geological strike ($N30^\circ E$).

10. The 3D inversion

In this study the inversion was done for the full static shift corrected impedance tensor (eight real and imaginary parts) of the 60 soundings shown in Fig. 15, and for 16 periods over 3.5 decades: i.e. 0.100, 0.251, 0.398, 0.631, 1.000, 1.585, 2.512, 3.981, 6.309, 10.000, 15.850, 25.120, 39.810, 63.090, 100.000 and 158.500 second periods. This amounts to a total of 7680 data values. The impedance values are weighted in the inversion by their measured variances. The five-layer model shown in Fig. 18 was used as the initial model. The inversion was run using Linux on an Intel Xeon 64-bit, 3.4 GHz processor. The memory in use was about 3.4 gigabytes and each iteration step took about 25 h. A total of 15 iterations were performed, taking almost 16 days (about 375 h) of continuous computer running time.

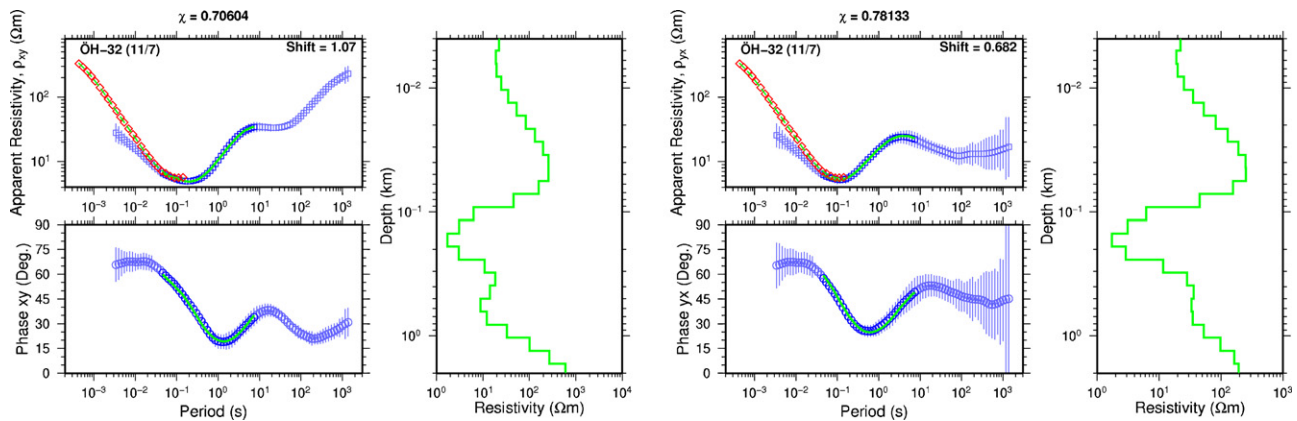


Fig. 16. Separate joint inversion of xy and yx modes of a MT sounding and a TEM sounding. Red diamonds: TEM apparent resistivities transformed to a pseudo-MT curve; blue squares: measured apparent resistivities; blue circles: apparent phase derived from the determinant of MT impedance tensor; light blue symbols to both sides of the green curves: data not used in the inversion; vertical blue lines: error bars; green lines in the two vertical panels: 1D resistivity inversion model; their synthetic MT apparent resistivity and phase response are shown in the panel to the left. The shift multiplier is shown in the upper right hand corner of the apparent resistivity panel. The number on the top of the figure (0581165): name of the MT station; bold number in the resistivity panels (OH-32): name of the TEM station. The number in parenthesis (11/7) means that the two stations were 11 m apart and their elevation difference was 7 m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

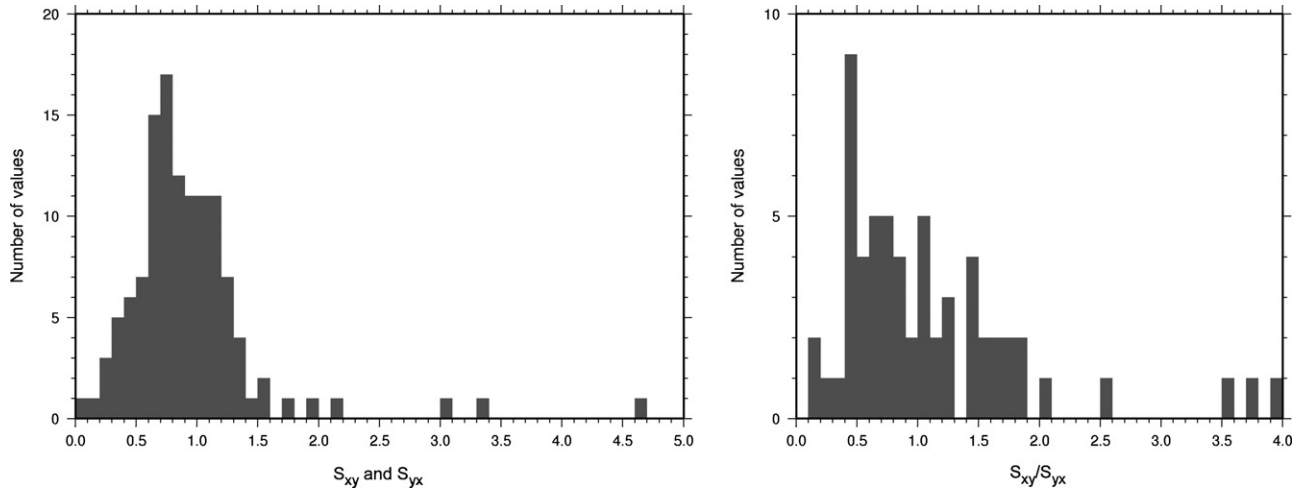


Fig. 17. Histogram of shift multipliers for the soundings used in the 3D inversion. Left panel: all the multipliers (both S_{xy} and S_{yx}). Right panel: their ratio (S_{xy}/S_{yx}).

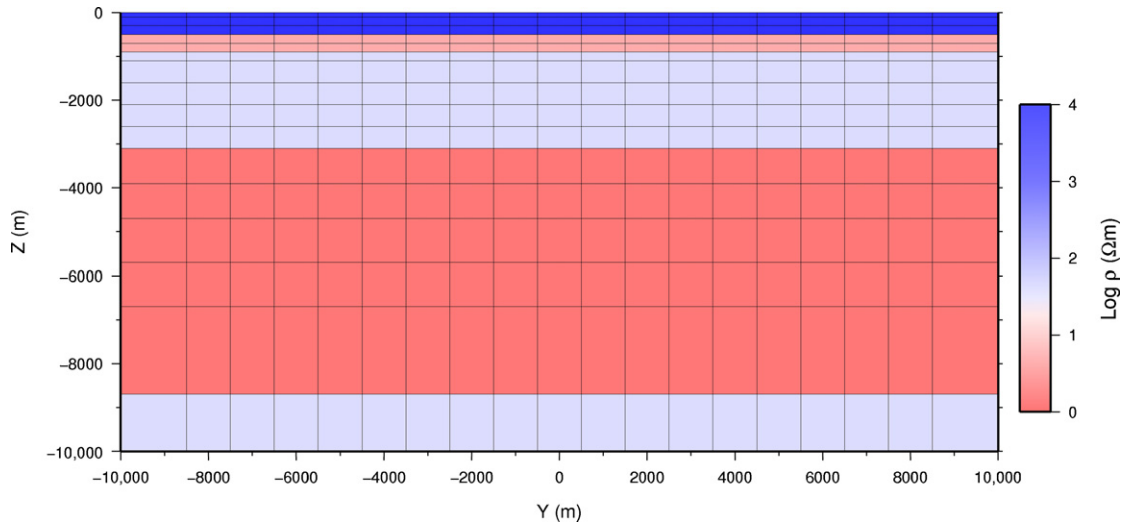


Fig. 18. Initial model for the 3D inversion of the MT data (central part). It is a five-layer model with a 500 Ω m surface layer, a shallow low-resistivity 10 Ω m layer, an intermediate high-resistivity 50 Ω m layer, a deep low-resistivity 5 Ω m layer and a 50 Ω m basement.

Fig. 19 displays the development of the root-mean-square (RMS) misfit during the inversion. The misfit drops sharply after the first iteration and more slowly during the next five. After that it stabilizes between 1.2 and 1.5, and subsequent iterations mainly produce a smoother model. Fig. 20 shows a typical development of the model response at four stages of the inversion computation and the measured sounding curves corrected for static shift. Model

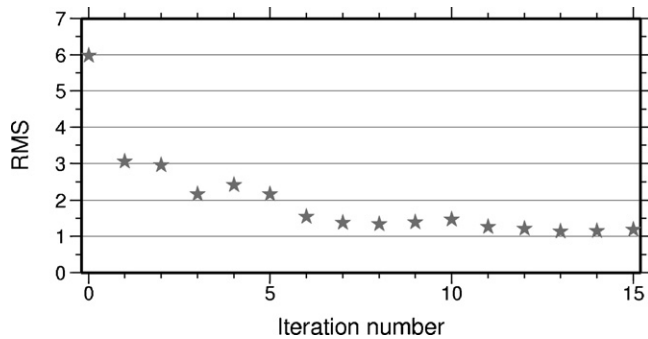


Fig. 19. Root-mean-square (RMS) misfit of the calculated and measured impedances (weighted by the variance) as a function of the iteration number.

calculations have demonstrated that at 10–20 km distance or more from the sea, which is the case for almost all the inverted soundings, its influence is expected to be negligible for the longest (158.5 s) periods used (Eysteinsson and Hermance, 1985).

11. Results of the inversion

The central part of the final model resulting from the 3D inversion is shown in Fig. 21(a)–(f), as smoothed resistivity maps at six different depths, corresponding to the centres of horizontal grid layers.

At shallow depth, 0.6 km below surface (Fig. 21(a)), the model shows low resistivity in a WNW–ESE elongated area running through and around Hengill and also in the SE part of the model. At this depth the resistivity is mainly controlled by the low-resistivity clay cap. A higher resistivity zone is starting to appear beneath and to the SE of Mount Hengill. However, the resolution at this shallow depth is poor because the shortest period used in the inversion is 0.1 s.

At 1.0 km depth (Fig. 21(b)) most of the areas that had low resistivity at shallow depth show high resistivity and in many places low resistivity is now observed where high resistivity was found at shallow depth. This reflects the spatial distribution of the depth

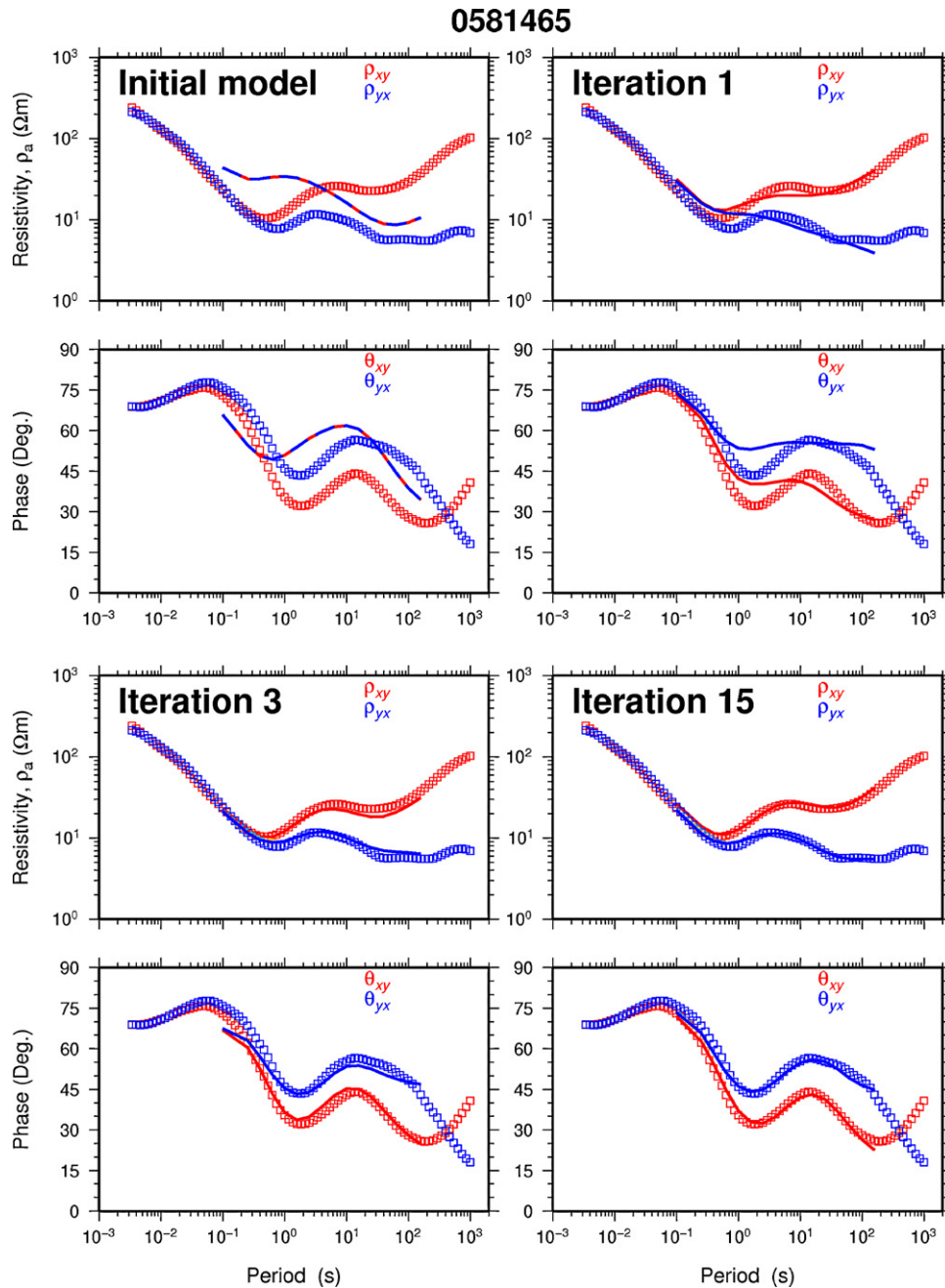


Fig. 20. Comparisons between measured values and model responses at four stages of the 3D inversion; i.e. for the initial model and after 1, 3 and 15 iterations. The measured values are static shift corrected observed apparent resistivity and phase. Red symbols: xy data; blue symbols: yx data; red lines: calculated responses for xy elements; blue lines: calculated responses for yx elements; red and blue lines in the initial model panels: calculated responses for the initial model. Number on the top of the figure (0581465): name of the MT station. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

to the conductive clay cap and to the resistive core. The resistivity variations at this depth are in general agreement with the depth to the resistive core as inferred from the 1D inversion (Fig. 14, left panel).

At 2.35 km depth (Fig. 21(c)), high resistivity is found in all the central region of the model. At 4.3 km (Fig. 21(d)), a deep low-resistivity zone starts to appear beneath Mount Hengill, extending SE towards the Grændalur central volcano. A low-resistivity region also extends to the west of Mount Hengill. At 6.2 km (Fig. 21(e)), the deep low-resistivity zone to the west of Mount Hengill spreads out towards the south and SE and low resistivity in the NW corner also becomes more prominent. The low-resistivity ridge to the SE of Hengill, bordered by high resistivity to the SW and NE, persists

and does not widen with depth. Below 9.7 km depth (Fig. 21(f)), the resistivity generally increases in the regions located under the deep low-resistivity zone.

As is seen in Figs. 21(a)–(f), the resistivity strike is mainly perpendicular to the geological strike (N30°E) and parallel to the y -axis in the rotated coordinate system of the model. Therefore, the central part of the model is shown as vertical cross-sections along the model x -axis (i.e. along the geological strike). Fig. 22 shows the location of the cross-sections (spaced 2 km apart and labelled by their y -coordinates in the model coordinate system). The sections are presented in Fig. 23(a)–(g).

All the sections show the shallow conductive clay cap with higher resistivity below but, as mentioned earlier, the resolution

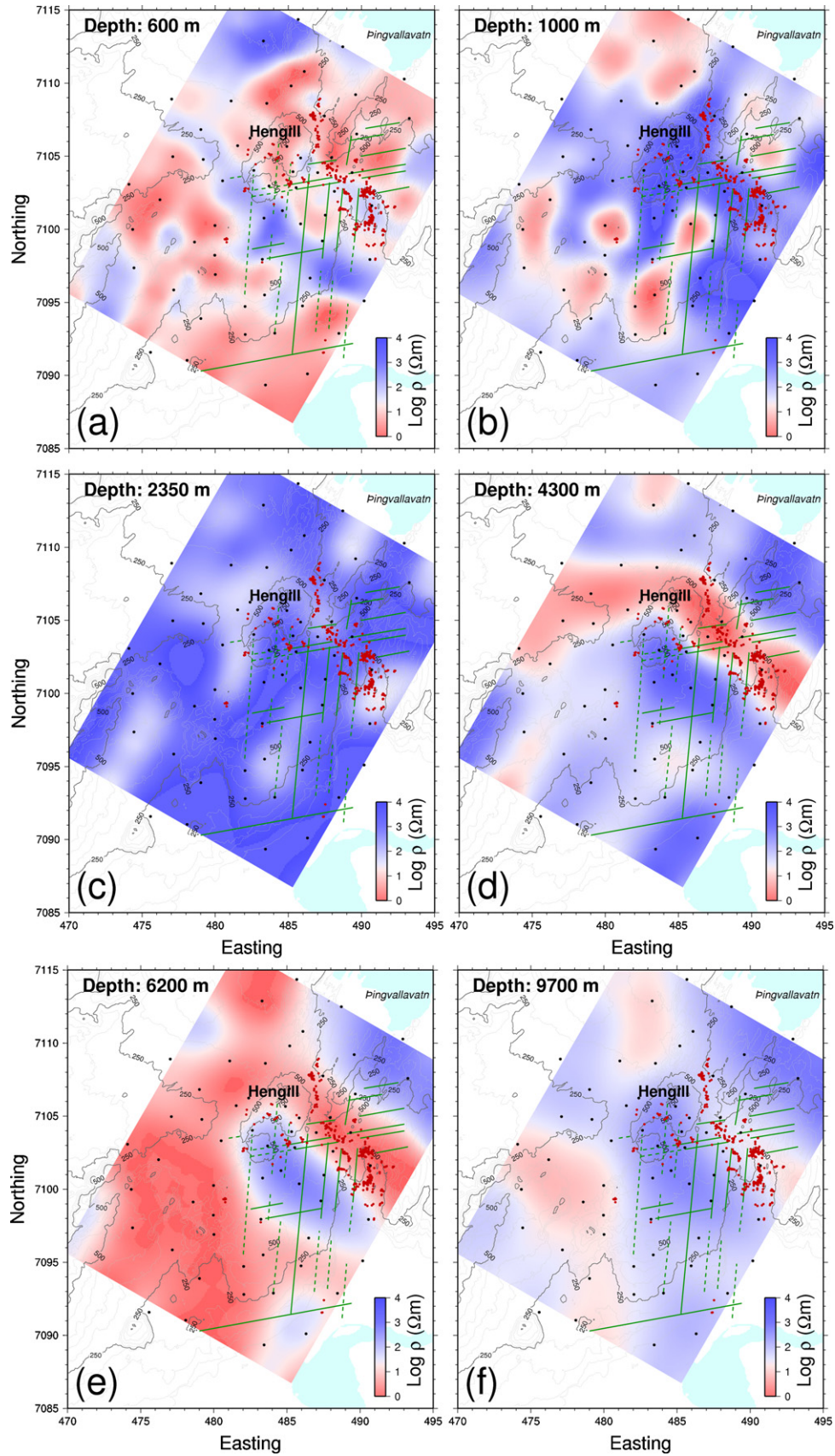


Fig. 21. Resistivity from 3D inversion for six depth slices; i.e. (a) at 0.6 km depth; (b) at 1.0 km depth; (c) at 2.35 km depth; (d) at 4.3 km depth; (e) at 6.2 km depth; and (f) at 9.7 km depth. Red dots: geothermal surface manifestations; black dots: MT soundings; green lines: faults inferred from seismic data; thin black lines: topographic contour lines in m a.s.l. Distances are given in km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

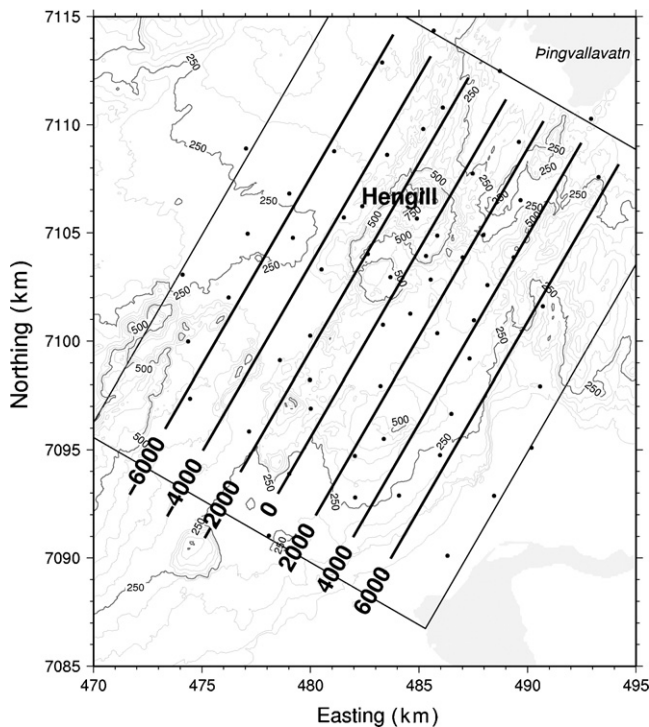


Fig. 22. Location of the cross-sections shown in Fig. 23(a)–(g). Black dots: MT soundings; thin black lines: topographic contour lines in m a.s.l. Distances are given in km.

is poor at shallow depth because only 0.1 s and longer periods are modelled. Modelling higher frequencies would have called for a model that included topography, which is not possible with the current version of the inversion code. The two sections farther to the NW (Fig. 23(a) and (b)) show continuous low resistivity at depth that domes up west of Mount Hengill. Farther to the SE (Fig. 23(c)–(g)) two deep conductive bodies are seen, one in the SW and another one beneath and to the SE of Mount Hengill. The top of this low-resistivity ridge is at about 3 km and extends to about 9 km depth. Its width is fairly constant at about 3.5 km.

12. Discussion of the 1D versus 3D results

The 3D inversion of the MT data in the Hengill area shows, in general, resistivity structures similar to the joint 1D inversion of the TEM and MT data. This correspondence between 1D and 3D results has been observed elsewhere (e.g. Cumming and Mackie, 2007). The elevation of the Hengill area is around 200–300 m, making it possible to compare Figs. 12 and 21 simply by adding 200–300 m to the depth scale in Fig. 21 and thereby comparing the results of the 1D and 3D inversions. The 3D inversion does, however, not give a detailed picture at shallow depth because the shortest period modelled in the 3D inversion is 0.1 s, and the areal coverage is not as good with only 60 soundings versus 148 soundings used in the 1D inversion.

Both inversion approaches show shallow low resistivity beneath and to the NW and SE of Mount Hengill that is underlain by higher resistivity (Fig. 12-top panels, and Fig. 21(a) and (b)). The two approaches also exhibit a deep NW–SE oriented conductor, at about 3 km depth, beneath and around Mount Hengill. At greater depth, at about 5–6 km, both approaches indicate low resistivity spreading out in the southern part of the Hengill area. At this depth there are, however, some differences between the models given by the 1D and 3D inversions.

The 3D inversion displays low resistivity below 5 km depth in the southern and western part of the mapped area (Fig. 21(e)), where the 1D inversion in some places shows high resistivity (Fig. 12, bottom right panel). Furthermore the 3D inversion indicates low resistivity extending farther to the SE from Hengill, towards the Grændalur complex, than the 1D inversion. The 3D model gives consistent high-resistivity SW of the low-resistivity zone located SE of Mount Hengill, where the 1D model yields somewhat lower resistivity (Fig. 23(e), (f) and Fig. 12-lower right panel).

Most of the MT soundings in the Hengill area show strong non-1D character, as illustrated by the divergence of the xy and yx apparent resistivity curves at periods over 1 s; see Fig. 20. The 3D inversion did successfully reproduce this character of the data, which lends more confidence to the 3D inversion results, compared to those given by the 1D model. However, although it may be surprising, it is impressive how well the 1D inversion resolves the major structures in the 3D model.

13. The nature of the deep conductors

As discussed earlier, the shallow resistivity structure of high-temperature geothermal systems in basaltic rocks is controlled by alteration mineralogy. On the other hand, the nature of the deep conductors is not as clear.

In the 1970s and 1980s, MT soundings showed that there is a conductive layer at 10–20 km depth under most of Iceland (Hermance, 1973; Beblo and Björnsson, 1978, 1980; Beblo et al., 1983; Hersir et al., 1984; Eysteinnsson and Hermance, 1985). This was initially interpreted to be due to partial melt in the crust. However, such a layer would be expected to attenuate seismic waves (especially S-waves) travelling through it. Based on a study of the attenuation of seismic waves in the Icelandic crust, strong arguments against the presence of partial melt at this depth interval have been put forward (Menke et al., 1995), at least for a partial melt that is thick or has more than a few percent liquid.

Recently, seismic activity that may be relevant to the discussion here, has been observed near Upptyppingar in the volcanic zone of northern Iceland (Jakobsdóttir et al., 2008), about 20 km east of the Askja central volcano. The activity is characterised by deep crustal seismicity, at 10–20 km depth, and shallower seismicity above 6–7 km depth. These two regions seem to be separated by an aseismic interval.

Two surveys of combined TEM and MT soundings have been conducted in the Upptyppingar area, which indicated a low-resistivity layer at about 2 km depth, underlain by high resistivity and a second low-resistivity layer at greater depth. The shallow low-resistivity layer reflects a conductive clay cap with 100–240 °C temperatures. The deep low-resistivity layer, observed in the about 7–13 km depth range (Vilhjálmsen et al., 2008), is part of one that extends under most of Iceland.

What is interesting is that the deep low-resistivity layer roughly coincides with the aseismic interval occurring between the deep and shallow seismicity regions. Furthermore, S-waves from deep-seated events travelled with little attenuation through the layer, implying that the deep low resistivity does not reflect extensive melt. The top of this layer coincides, however, roughly with increasing P-wave velocity as determined from refraction seismics (Pálmason, 1971).

A recent combined TEM and MT survey in the Krafla geothermal area revealed deep conductors that dome up to 2.5–3 km depth in the volcanic complex (Árnason et al., 2009). The Krafla volcano was very active in 1975–1984 with several rifting episodes, dike injections and eruptions. Analyses of earthquake data recorded by a local seismic network revealed that S-waves could not penetrate through, or were highly attenuated in some parts of the Krafla vol-

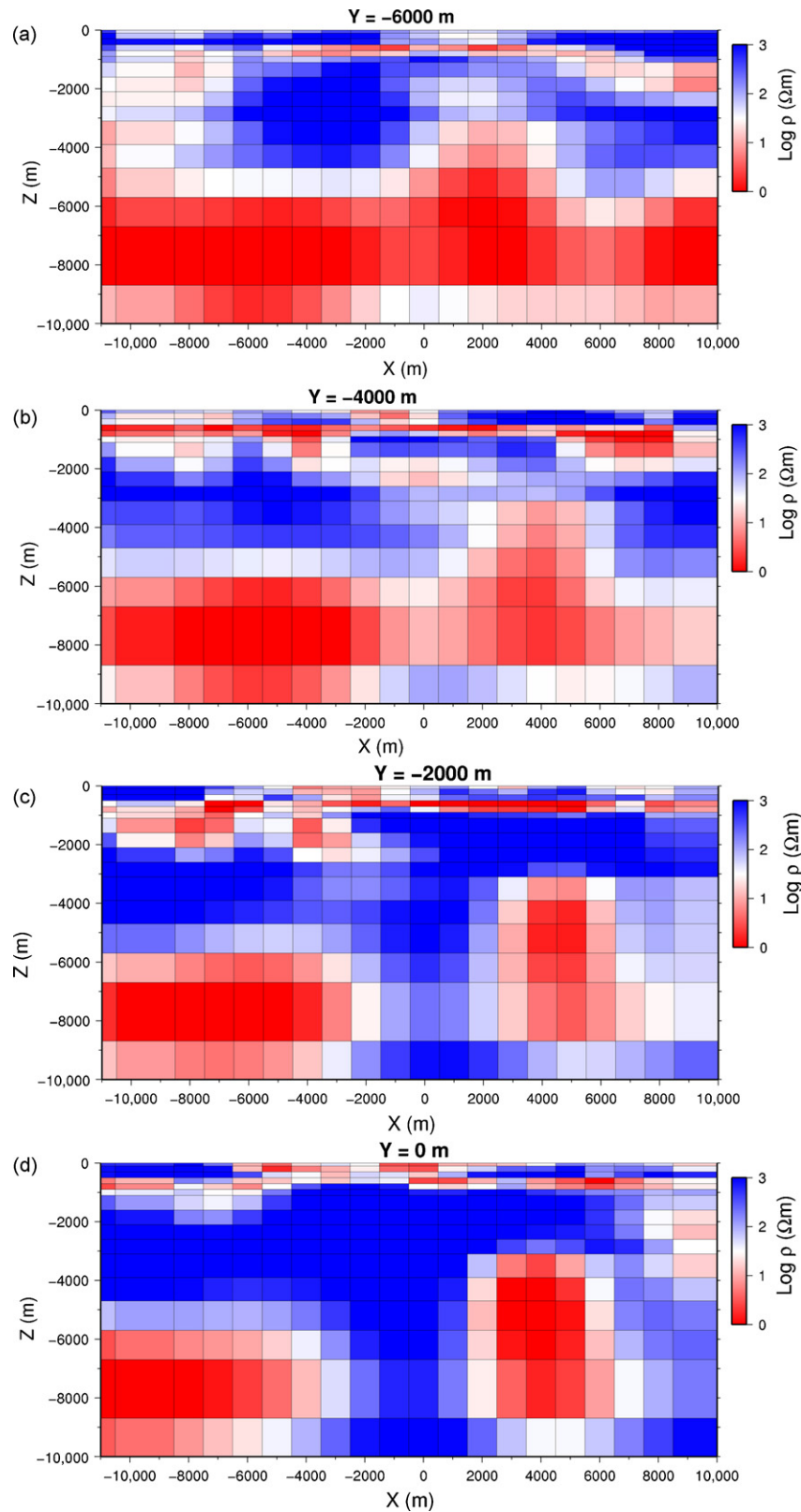


Fig. 23. (a) SW–NE inverted resistivity section across the model at $y = -6000 \text{ m}$ (for location see Fig. 22). (b) SW–NE inverted resistivity section across the model at $y = -4000 \text{ m}$ (for location see Fig. 22). (c) SW–NE inverted resistivity section across the model at $y = -2000 \text{ m}$ (for location see Fig. 22). (d) SW–NE inverted resistivity section across the model at $y = 0 \text{ m}$ (for location see Fig. 22). (e) SW–NE inverted resistivity section across the model at $y = 2000 \text{ m}$ (for location see Fig. 22). (f) SW–NE inverted resistivity section across the model at $y = 4000 \text{ m}$ (for location see Fig. 22). (g) SW–NE inverted resistivity section across the model at $y = 6000 \text{ m}$ (for location see Fig. 22). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

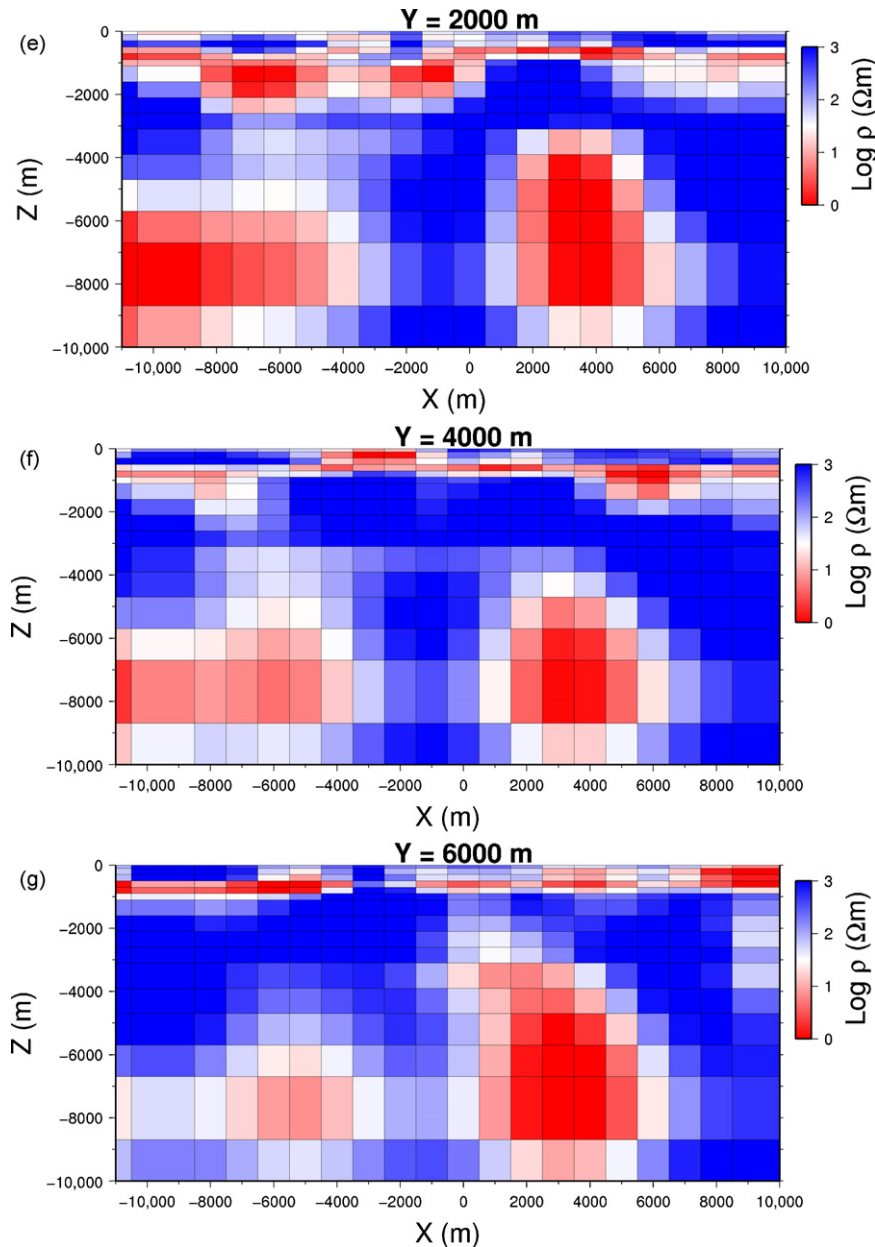


Fig. 23. (Continued).

cano. This was interpreted as indicating the existence of magma chambers with molten rocks at 3–7 km depth (Einarsson, 1978). Since the location and depth of the S-wave attenuation coincides with the up-doming deep conductors seen in the TEM/MT survey, it was concluded that they reflect molten rocks.

There is no evidence for persistent magma chamber(s) under the Hengill area, except maybe for the uplift revealed by the InSAR data discussed earlier (Feigl et al., 2000). The uplift was, however, localized over only a small part of the deep conductors. Analyses of seismic data have not shown any convincing signs of S-wave attenuation under the study area. To the contrary, seismic tomography has indicated a low V_p/V_s ratio at depth (Tryggvason et al., 2002; Jousset et al., 2010). We, therefore, think that the nature of the deep conductors in the Hengill area is like that of the deep low-resistivity layer observed at Upptyppingar.

The crust under Iceland is considered to be similar to an oceanic type, but much thicker (Pálmason, 1971). The typical oceanic crust has been studied in ophiolites, which are fragments of oceanic crust

that have been pushed up onto continental crust (Dilek et al., 2000). The normal oceanic crust is about 5–10 km thick, bounded below by the Moho discontinuity at the top of the mantle. It consists of three main layers. The lowermost consists of gabbroic rocks that are thought to be left behind by differentiation of mantle material in magma chambers. The lighter magma migrates up into the second layer, which is a dike complex (sheeted dikes). It either creates dikes or reaches the surface, producing the uppermost part of the oceanic crust that is composed of volcanic rocks.

As stated earlier, the Icelandic crust is generally thought to be much thicker than the normal oceanic crust, or 20–40 km (Bjarnason, 2008; Brandsdóttir and Menke, 2008). It is believed that the extrusive part is about 6–7 km thick (Kristján Sæmundsson, pers. comm., November 2009). This would correspond to the brittle upper part beneath Upptyppingar, with the conductive clay cap and the resistive layer below it. We propose that the deep low-resistivity and aseismic layer at 7–13 km depth most likely reflects the sheeted dike complex, and that the more resistive layer below

corresponds to the gabbroic rocks. The basaltic dikes and intrusions are ductile, while the gabbros below are more brittle. The ductile–brittle transition in basalts takes place at about 600–700 °C (Warren and Latham, 1970). The base of the upper brittle part of the crust beneath Upptyppingar, which coincides with the top of the deep low-resistivity layer, therefore probably corresponds to an isotherm close to 600–700 °C.

An important point needs to be addressed. If partial melt is not present, why are the ductile rocks conductive? Fournier (1999) has suggested that below the ductile–brittle boundary, magmatic water is trapped within the plastic rocks. Chemical species that do not fit into the mineral structures when the magma solidifies are dissolved in the remaining water. In chlorine-rich acidic rocks this can lead to concentrated brine because under exceptionally high strain rates, the rocks hosting the water can break and the consequent pressure release (from lithostatic to hydrostatic) causes boiling and escape of supercritical “steam”. The dissolved ions probably stay behind increasing their concentration in the remaining brine. Basaltic magma is much lower in chlorine than acidic magma. It is, however, possible that segregation by the just described boiling process is efficient enough to eventually produce conductive brines in basaltic rocks. Although the reason for finding localised deep conductors in the crust, like in the Hengill area, is not clear, they can, with relatively good confidence, be assumed to be closely related to the heat sources of hydrothermal systems.

The deep conducting layer under the Upptyppingar area seems to be almost horizontal. This is in general agreement with early studies (see above) and probably reflects the general deep resistivity structure of most of the crust beneath Iceland, outside the central volcanoes. It is evident from the 3D (and the 1D) model discussed earlier, that the resistivity structure beneath the Hengill central volcano is much more complex.

The most dominant feature of the resistivity model is a shallow and narrow low-resistivity ridge extending from NW of Mount Hengill to the SE, towards Grændalur. It can be seen (Fig. 21(d) and (e)) that this anomaly coincides with an area of “wedges”, where E–W transform faults meet N–S oriented faults. As previ-

ously discussed, fault movement activity generally started on the E–W faults and then was taken over by the N–S faults with transform crustal movement. This “wedge” formation by the transform tectonics might facilitate magma intrusions and the emplacement of igneous dikes.

Fig. 24 (left panel) shows a residual Bouguer gravity map of the Hengill area (Porbergsson et al., 1984) and a comparison with resistivity at 5.2 km depth (Fig. 24, right panel). It shows that the low-resistivity anomaly SE of Mount Hengill coincides with a gravity high and that there seems to be a general correlation between high gravity values and low resistivity at depth. This lends further support to the interpretation that the deep conductors represent dense igneous intrusions and dikes. Drilling in Nesjavellir, just north of Mount Hengill, showed that below about 2 km depth, around 80% of the rocks are intrusives (Franzson, 1998) and recent drilling south of Hengill has shown similar character (Hjalti Franzson, pers. comm., November 2009).

There is an indication (Fig. 3) that another NW–SE oriented zone of “wedges” formed by E–W and N–S transform faults exists about 7 km south of the zone going through Hengill. This zone is underlain by a deep conductor. Between these zones and SE of the fissure swarm, the 3D resistivity model shows no deep conductors, and the gravity map indicates relatively low gravity values. This in-between area corresponds to a plateau of post-glacial lavas, while the zones of the wedges to the NE and SW are characterised by higher elevations due to more extensive (mostly sub-glacial) volcanism.

As mentioned earlier, a broadband passive seismic survey was done in the Hengill area as part of the I-GET project (Jousset et al., 2010). Seven broadband seismometers were deployed in the summer of 2006 and data were recorded for about 3.5 months. A tomography inversion was performed for recorded events with clear P- and S-wave picks (total of 260). The tomography inversion showed a high P-wave velocity anomaly at 2–4 km depth SE of Mount Hengill.

Most of the seismicity recorded by the broadband network was to the SE of Mount Hengill. The depth of the events was mainly between 2 and 5 km (Jousset et al., 2010), that is above or in the

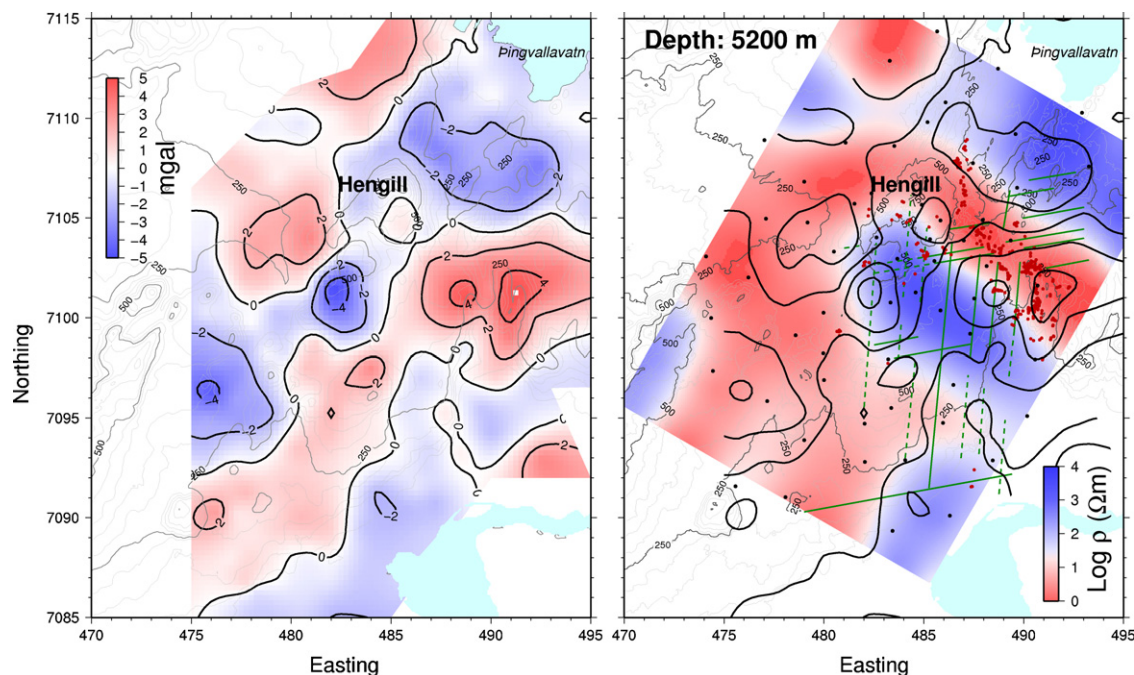


Fig. 24. (Left) Residual Bouguer gravity map of the Hengill area; (right) comparison of gravity anomalies, shown by the black contours, and resistivity at 5.2 km depth. Red dots: geothermal surface manifestations; black dots: MT soundings; green lines: faults inferred from seismic data; thin black lines: topographic contour lines in m a.s.l. Distances are given in km. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

upper part of the deep conductor. This is also similar to what was observed at Upptyppingar, except that no deep seismicity was detected beneath the Hengill area. The depth of the bulk of the seismic events recorded by the permanent (SIL) seismic network in the period from 1991 to 2001 and located by the Icelandic Meteorological Office (IMO) is in the 2–7 km range, with a peak at about 5 km. It has, however, become apparent that the velocity model used for locating the earthquakes is inadequate and results in events locations that are too deep. A subset of the earthquakes has been relocated with an improved velocity model, giving similar depth range as in the broadband experiment, i.e. 2–5 km (Vogfjörð and Hjaltadóttir, 2007; Hjaltadóttir and Vogfjörð, 2009).

We, therefore, believe that the deep conductors in the Hengill and Upptyppingar areas reflect dikes and magma intrusions. At Upptyppingar, which is within the volcanic rift zone, but not inside a central volcano, a fairly horizontal deep conducting layer is found at 7–13 km depth. At Hengill the distribution of the deep conductor is much more complex. Its structure correlates with the transform tectonics that was active during the 1991–2001 period, which showed first movements on the E–W faults and later on the N–S faults, and may have facilitated magma intrusions into the crust. The uplift SE of Mount Hengill during the period of intense seismicity, revealed by the InSAR data (Feigl et al., 2000), probably reflects the intrusions initiated by the transform movements.

As discussed before, there are no convincing signs of S-wave attenuation under the Hengill area, and, therefore, we believe that there is no extensive magma chamber in the Hengill volcanic complex. This implies that the volcanic system in the area is less developed than the one at Krafla, where all evidence points to the presence of shallow magma chamber(s). The crustal structure at Hengill is therefore more like that beneath Upptyppingar, but due to the transform tectonics interacting with the crustal spreading, magma injection into the crust is facilitated, forming dikes and intrusions at shallow depth, as well as volcanic eruptions.

The reason for having a thicker crust under Iceland, than under the oceanic ridges, is magma production due to the mantle plume (hotspot) present under Iceland. The plume is moving eastwards relative to the Mid-Atlantic Ridge and diverts the crustal spreading in Iceland to the east of the oceanic ridge SW and north of Iceland. On macro scale, this is adjusted by big transform zones in the South Iceland Seismic Zone (SISZ), and in the Tjörnes–Flatey transform zone, which is located in northern Iceland, and north of the island. In between, the spreading takes place in NE–SW and N–S trending volcanic zones. The growth of volcanic zones adjusts to the general WNW–ESE spreading of the Mid-Atlantic Ridge, by way of transform tectonics. An example would be the Hengill–Hrómundartindur–Grændalur volcanic complex, which is located about where transform tectonics interferes with crustal spreading (i.e. at the western end of the SISZ). It is tempting to suggest that the main volcanic centres (central volcanoes) in Iceland are formed where the spreading is adjusted by transform tectonics.

This and other recent studies of the Hengill area (e.g. InSAR) indicate that the formation of shallow intrusions (at 3–6 km depth) is most intense under and to the SE of Mount Hengill. These intrusions are heat sources for the overlying geothermal system. But as mentioned earlier, well ÖJ-1 drilled SE of Hengill (Fig. 2) shows a nearly constant temperature of only 200 °C between 500 and 1000 m depth (Fig. 6), well below the boiling point curve. This seems to contradict the conclusion that intensive heat sources exist at a shallow (3–4 km) depth in this area. In parts of the Krafla geothermal system two thermal regimes are observed. In its upper part, similar temperatures as in ÖJ-1 (nearly constant 200 °C) are found (Ármansson et al., 1987). Below that, downhole logs exhibit a steep linear increase of temperature with depth, reflecting conductive heat transfer. Beneath this thermally conductive zone, the temperature follows the boiling point curve. In other words, there

are two geothermal reservoirs at Krafla, an upper one of about 200 °C and a deeper one with much higher temperatures, separated by an impermeable zone that conducts the heat from the lower to the upper reservoir. This zone is found where the rocks change from extrusive to dominantly intrusive. We, therefore, suggest that a similar situation prevails SE of Mount Hengill, i.e. there is an upper reservoir located above impermeable intrusions and a much hotter reservoir at greater depths, as in Krafla.

14. Conclusions

This study demonstrates that the combined use of TEM and MT soundings is a very powerful tool for studying geothermal systems in Icelandic volcanic environments from near surface to great depths. In such environments, one can expect very high near-surface resistivity contrasts leading to severe and often systematic static shifts over relatively large regions. Therefore, static shift correction in MT soundings is necessary and a joint inversion of TEM and MT data is a consistent way to accomplish this. A combination of MT and TEM has been used successfully in other areas (e.g. Anderson et al., 2000; Cumming et al., 2000).

Joint inversion of TEM and rotationally invariant determinant MT data from 148 sounding sites in the Hengill area reveals a resistivity structure consisting of a shallow low-resistivity layer, in about the uppermost 2 km, underlain by high resistivity. At greater depth, a second low-resistivity layer is observed in most of the area, again underlain by higher resistivity. The depth to this second layer varies over the study area. It is at shallowest depth (about 3 km) under and around Mount Hengill. The nature of the upper low-resistivity layer is known by comparing results from DC and TEM soundings and borehole data; it reflects conductive hydrothermal alteration minerals formed at temperatures between 100 and 240 °C. The nature of the deep conductive layer is not as clear, but its high conductivity could be due to magmatic brines trapped in ductile intrusive rocks.

A full 3D inversion of the static shift corrected MT tensor for 60 soundings gives results similar to the 1D inversion, but a sharper picture of the subsurface is obtained; as expected, the 1D inversion smears out resistivity anomalies. The 3D inversion shows deep low resistivity under most, but not all, of the survey area. A NW–SE oriented low-resistivity anomaly through Mount Hengill and SE of it is seen. This anomaly is about 3.5 km wide and extends from about 3–9 km depth. Farther to the SW another NW–SE oriented zone of low resistivity is observed at somewhat greater depth. Southeast of the fissure swarm they are separated by high resistivity, but NW of it the two zones are connected. The low-resistivity anomalies at depth correlate with relatively positive residual Bouguer gravity, implying higher density.

The NW–SE oriented, low-resistivity anomaly at 3–9 km under and to the SE of Mount Hengill is found where intense seismic activity associated with transform tectonics occurs. Since no attenuation of S-waves is observed under the Hengill area, the deep conductors are believed to reflect hot, solidified intrusions that are heat sources for the geothermal system above.

We conclude from the present study that a multi-method approach with joint interpretation of TEM, MT, seismic and gravity data, and the use of advanced interpretation techniques can reveal the structure and nature of high-temperature geothermal systems in volcanic environments, and shed light on the nature of the tectonics and volcanism of the area being studied.

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